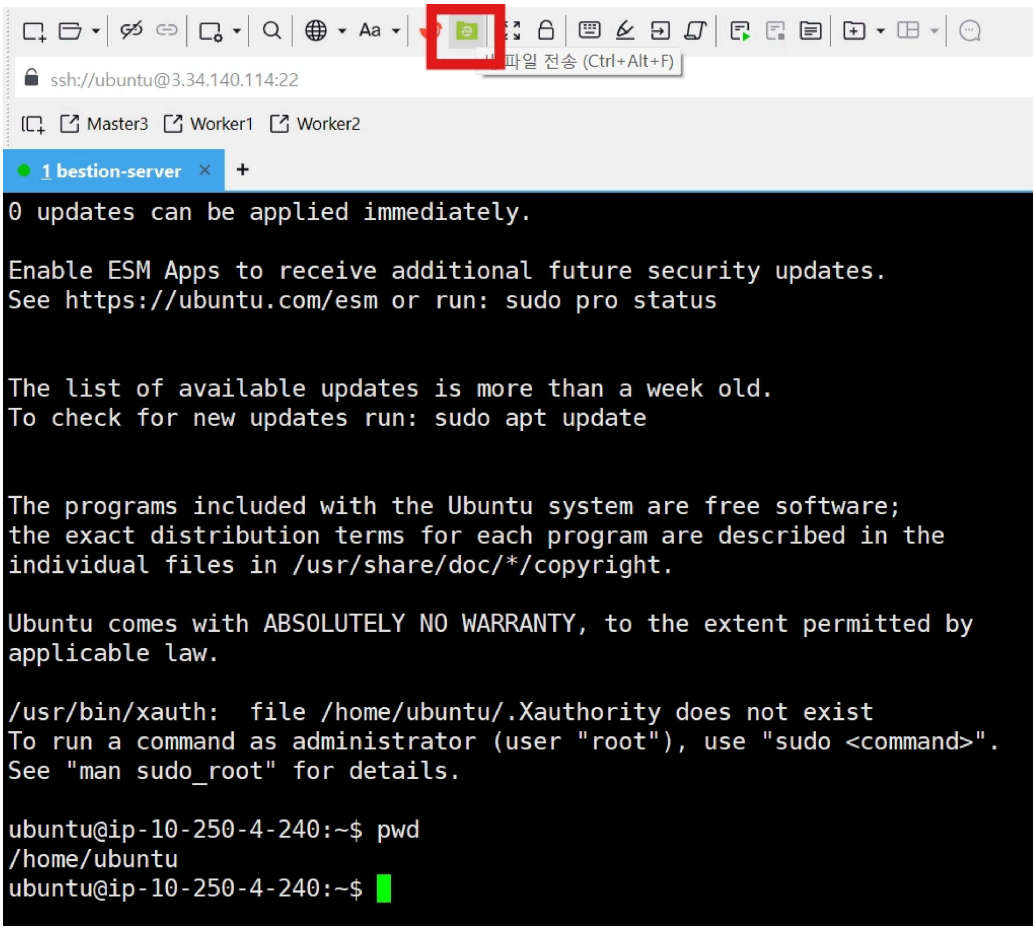
//

[10]. AWS EC2 환경에서 3Tier 통신을 통한 웹서비스 구축(핵심)

생성일

태그

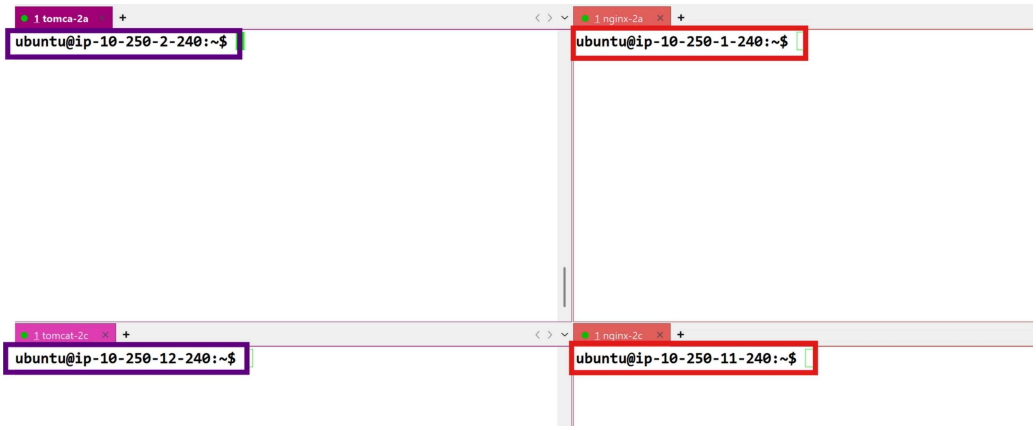
01.Bastion-Server 통해 nginx2a/nginx2c 으로 접속



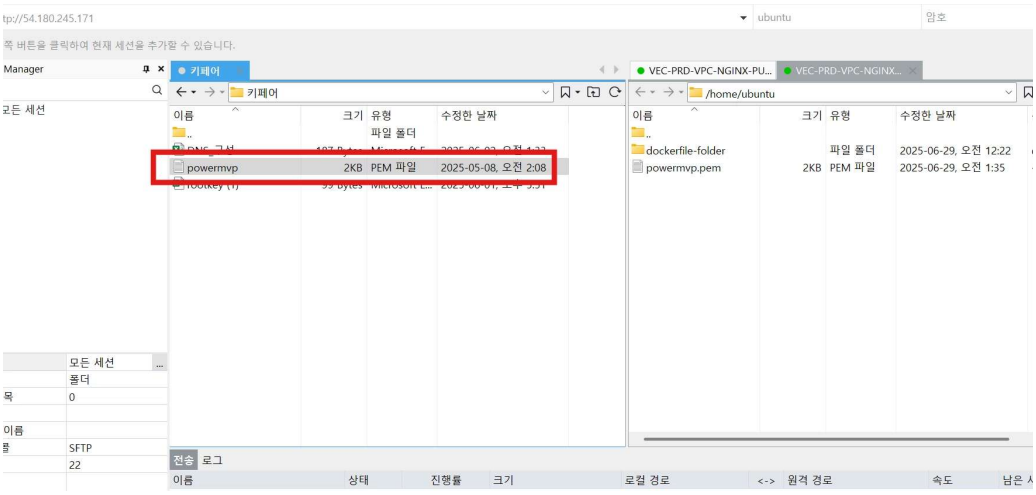
이 상태에서 아래와 같은 명령어를 통해 Tomcat이나 Nginx에 접속하여 사용한다.

1. 단, 사전에 Bastion-Server 가 접속하고자하는 서버의 **powermvp.pem** 키에 대한 복사 필요
2. 사전에 인바운드 보안정책에서 Port 22 개방 필요

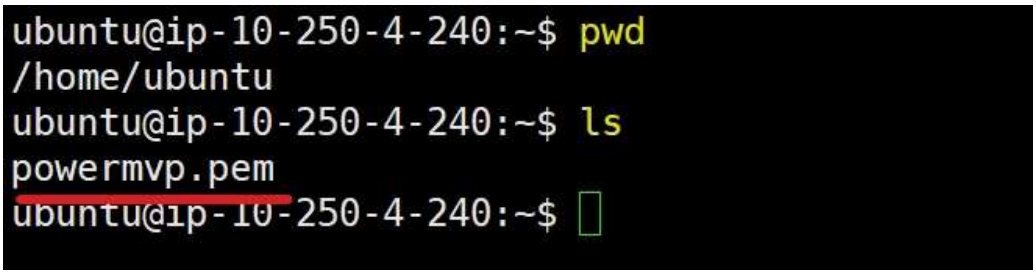
```
ssh -i powermvp.pem ubuntu@10.250.1.240
ssh -i powermvp.pem ubuntu@10.250.11.240
ssh -i powermvp.pem ubuntu@10.250.2.240
ssh -i powermvp.pem ubuntu@10.250.12.240
```



사전에 먼저 Bastion-Server에서 아래 작업을 진행한다.



SFTP로 생성한 pem를 이동시키기 (로컬서버 → Bastion EC2로 이동)



펨키를 사용하여 원격접속을 할 수 있음

```
chmod 400 (자신의 펨키 이름)
```

```
ssh -i (자신의 펨키 이름) ubuntu@10.250.1.240
```

AWS 에서 일반적인 서버간 연결 방법

→ **ssh -i powermvp.pem ubuntu@10.250.1.240 (Ubuntu Linux계정)**

→ **ssh -i powermvp.pem ec2-user@10.250.1.240 (AWS Linux계정)**

```
ubuntu@ip-10-250-4-240:~$ chmod 400 powermvp.pem
ubuntu@ip-10-250-4-240:~$ ssh -i powermvp.pem ubuntu@10.250.1.240
The authenticity of host '10.250.1.240 (10.250.1.240)' can't be established.
ED25519 key fingerprint is SHA256:KjZJDiv3MEkJdZv+2183GL6Ab8WOBeVOI8usloGuxrA.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.250.1.240' (ED25519) to the list of known hosts.
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-1029-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Thu Aug 14 09:27:12 UTC 2025

  System load:  0.0               Processes:             102
  Usage of /:   22.0% of 7.57GB   Users logged in:       0
  Memory usage: 23%               IPv4 address for ens5: 10.250.1.240
  Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.
```

```
exit
```

```
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-10-250-1-240:~$ exit
logout
Connection to 10.250.1.240 closed.
ubuntu@ip-10-250-4-240:~$
```

사용에 편의성을 위해 config 설정을 통해 서버간 간편 연결 방법 (참고사항)

```
vi ~/.ssh/config
```

파일 새로 생성 후 아래 내용 붙여넣기

```
Host nginx2a
    HostName 10.250.1.240
    User ubuntu
    IdentityFile ~/.ssh/powermvp.pem
    StrictHostKeyChecking no
    UserKnownHostsFile /dev/null

Host nginx2c
    HostName 10.250.11.240
```

```
    User ubuntu
    IdentityFile ~/.ssh/powermvp.pem
    StrictHostKeyChecking no
    UserKnownHostsFile /dev/null

Host tomcat2a
    HostName 10.250.2.240
    User ubuntu
    IdentityFile ~/.ssh/powermvp.pem
    StrictHostKeyChecking no
    UserKnownHostsFile /dev/null

Host tomcat2c
    HostName 10.250.12.240
    User ubuntu
    IdentityFile ~/.ssh/powermvp.pem
    StrictHostKeyChecking no
    UserKnownHostsFile /dev/null
```

```
Host nginx2a
     HostName 10.250.1.240
     User ubuntu
     IdentityFile ~/.ssh/powermvp.pem
     StrictHostKeyChecking no
     UserKnownHostsFile /dev/null

Host nginx2c
     HostName 10.250.11.240
     User ubuntu
     IdentityFile ~/.ssh/powermvp.pem
     StrictHostKeyChecking no
     UserKnownHostsFile /dev/null

Host tomcat2a
     HostName 10.250.2.240
     User ubuntu
     IdentityFile ~/.ssh/powermvp.pem
     StrictHostKeyChecking no
     UserKnownHostsFile /dev/null

Host tomcat2c
     HostName 10.250.12.240
     User ubuntu
     IdentityFile ~/.ssh/powermvp.pem
     StrictHostKeyChecking no
     UserKnownHostsFile /dev/null
```

자신의 펨키를 안전하게 .ssh/ 디렉터리로 이동

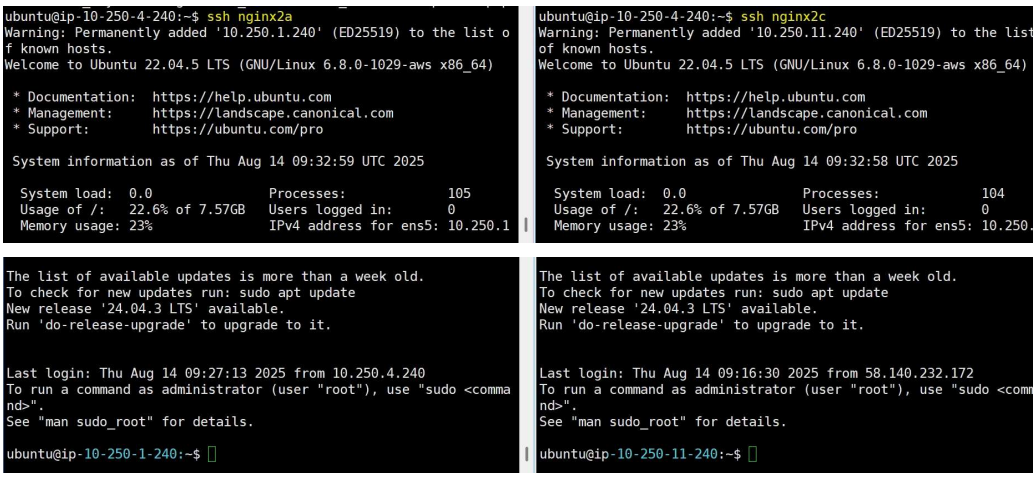
```
mv powermvp.pem ~/.ssh/
```

```
ubuntu@ip-10-250-4-240:~$ ls
powermvp.pem
ubuntu@ip-10-250-4-240:~$ mv powermvp.pem ~/.ssh/
ubuntu@ip-10-250-4-240:~$ ls
ubuntu@ip-10-250-4-240:~$ ls ~/.ssh/
authorized_keys  config  known_hosts  known_hosts.old  powermvp.pem
```

config 설정을 마쳤으므로 아래 명령어로 WEB EC2로 접속

```
ssh nginx2a
```

```
ssh nginx2c
```



02. Bastion-Server 접속후 NGINX 설치 (nginx-2a, nginx-2c 모두에서 진행)

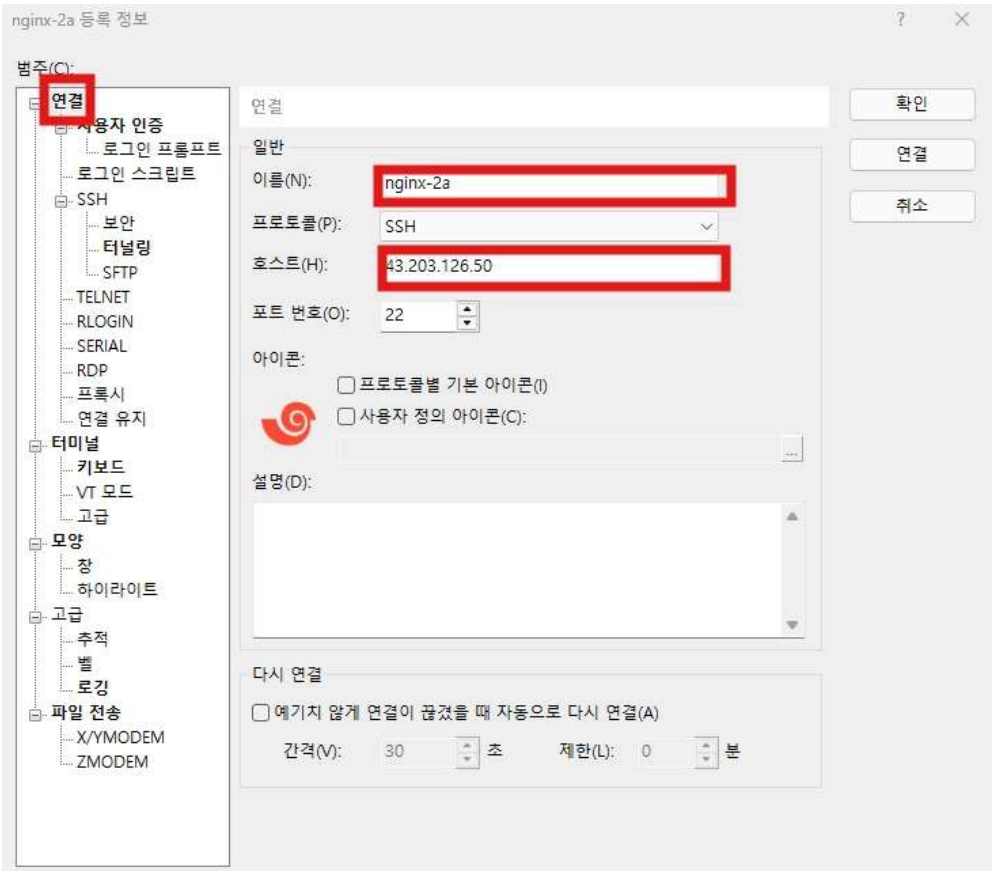
1) XShell 설치 및 SSH 접속

XShell 설치경로 : https://www.netsarang.com/ko/xshell-download/

설치 후 아래와 같이 작성 → 순서를 위해 Tomcat 서버와 연결

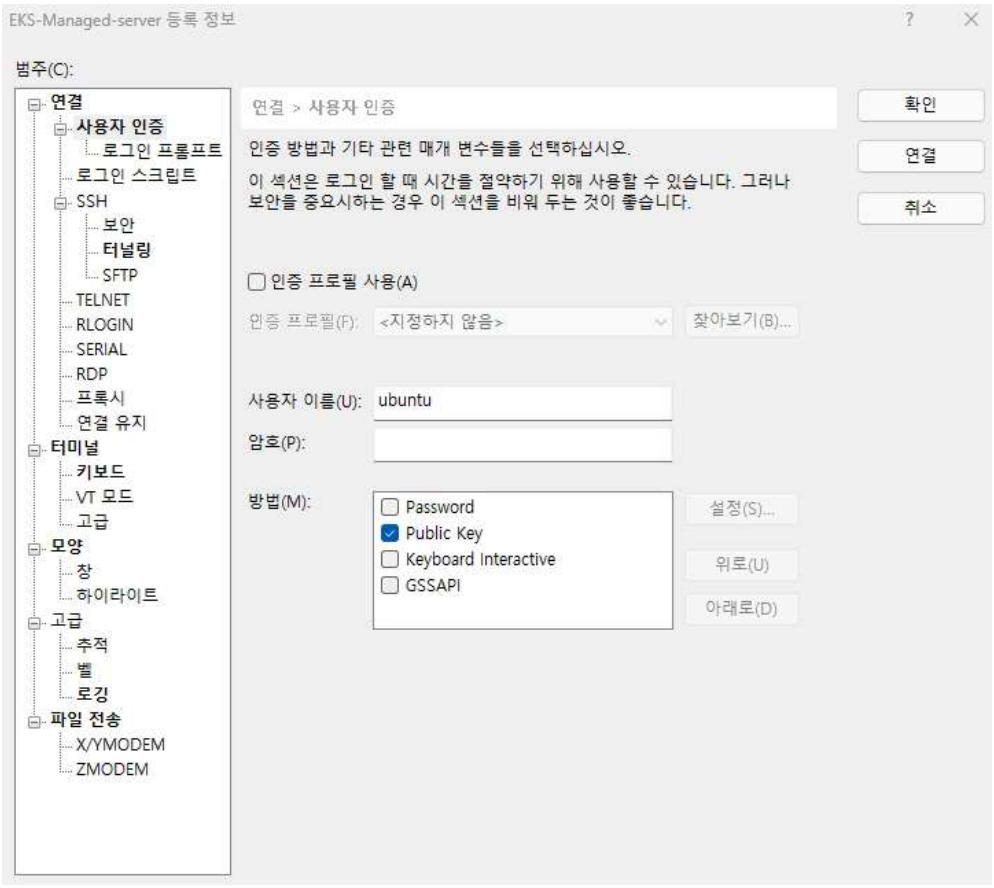
연결

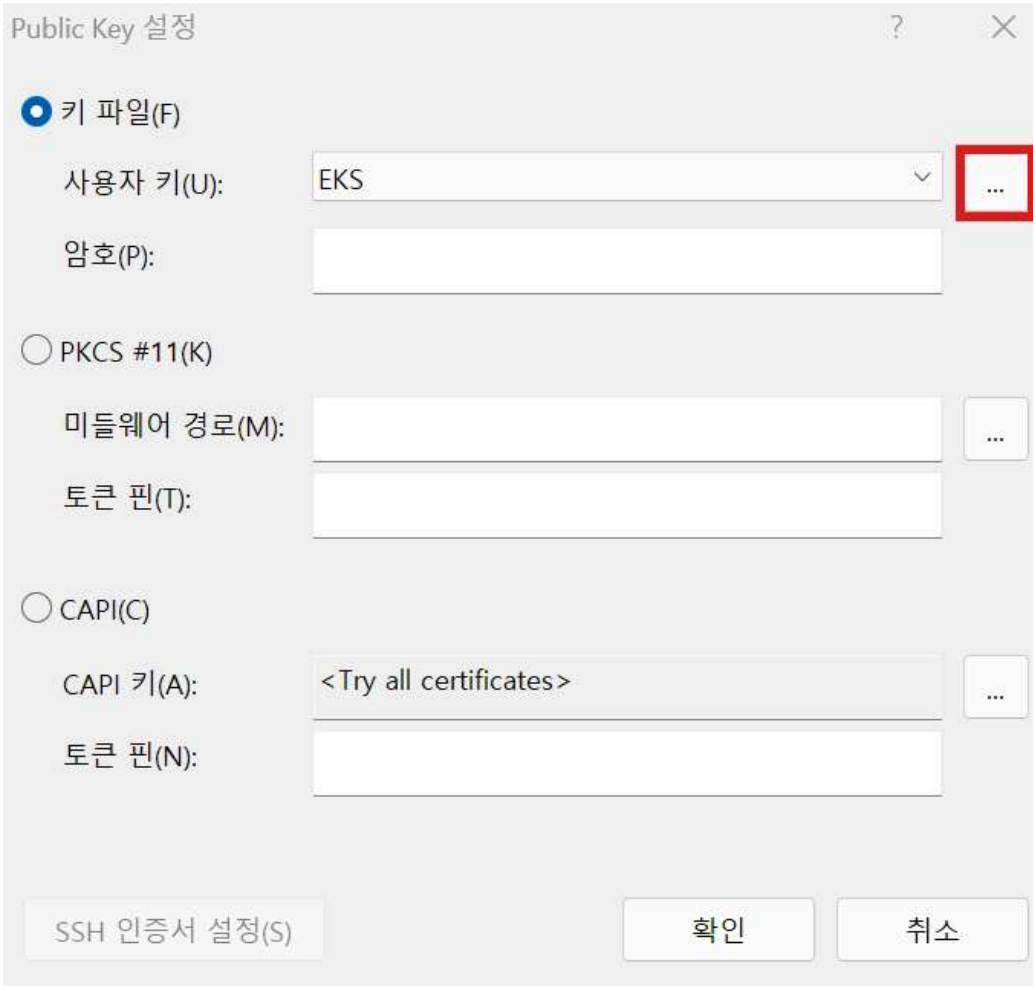
이름	nginx-2a
호스트	(퍼블릭 IPv4 주소)
포트 번호	22



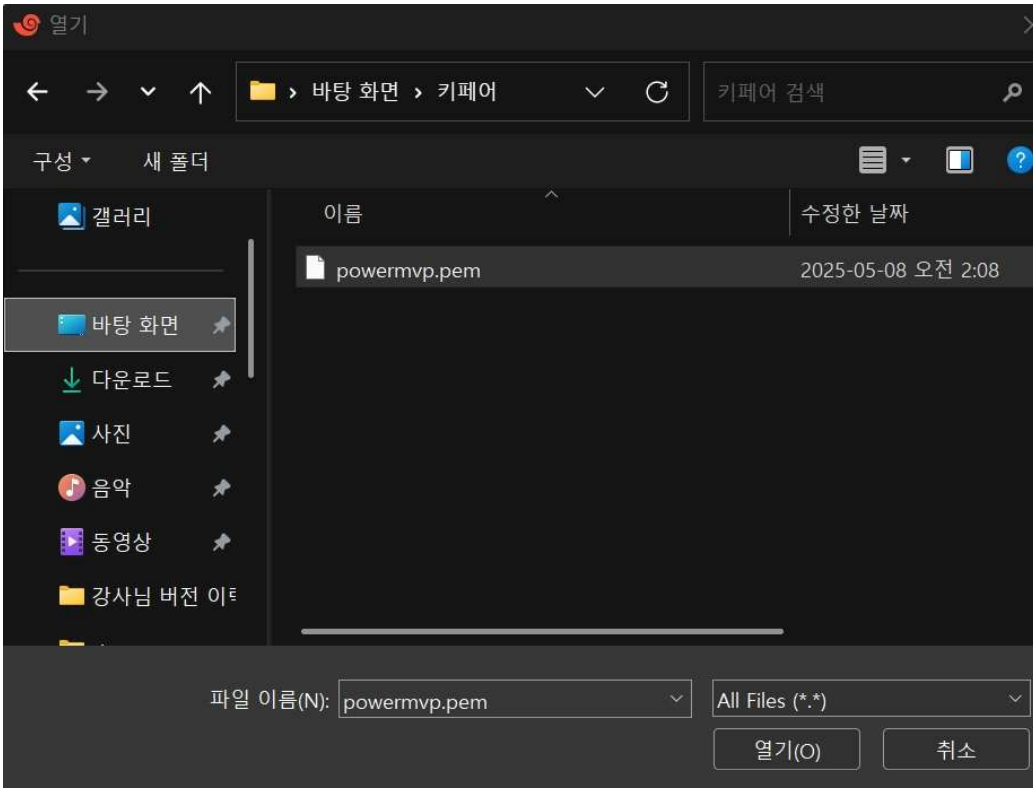
사용자 인증

사용자 이름	ubuntu
방법	Public key
키파일	(생성한 키페어)

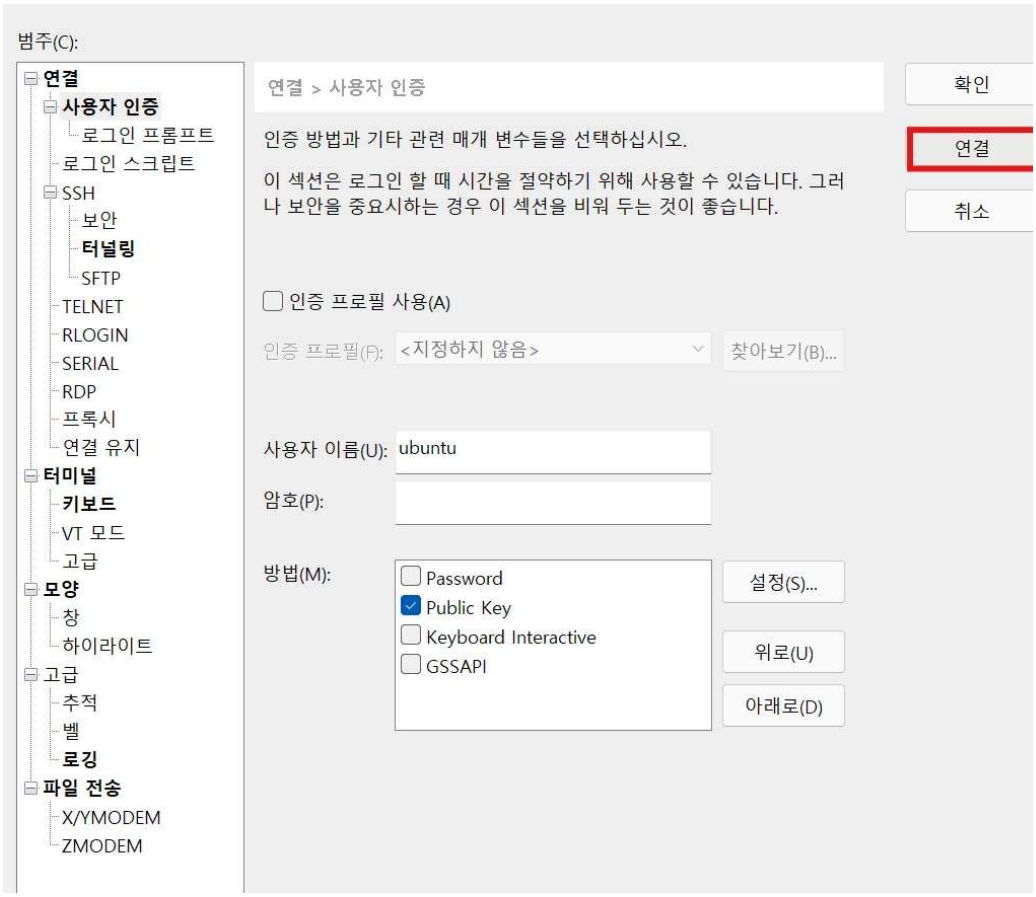

EKS-Managed-server 등록 정보
범주(C):
연결
사용자 인증
로그인 프롬프트
로그인 스크립트
SSH
보안
터널링
SFTP
TELNET
RLOGIN
SERIAL
RDP
프록시
연결 유지
터미널
키보드
VT 모드
고급
모양
창
하이라이트
고급
추적
벨
로깅
파일 전송
X/YMODEM
ZMODEM
연결 > 사용자 인증
인증 방법과 기타 관련 매개 변수들을 선택하십시오.
이 섹션은 로그인 할 때 시간을 절약하기 위해 사용할 수 있습니다. 그러나 보안을 중요시하는 경우 이 섹션을 비워 두는 것이 좋습니다.
확인
연결
취소
인증 프로필 사용(A)
인증 프로필(F):
<지정하지 않음>
찾아보기(B)...
사용자 이름(U):
ubuntu
암호(P):
방법(M):
Password
Public Key
Keyboard Interactive
GSSAPI
설정(S)...
위로(U)
아래로(D)


Public Key 설정
키 파일(F)
사용자 키(U):
EKS
...
암호(P):
PKCS #11(K)
미들웨어 경로(M):
...
토큰 핀(T):
CAPI(C)
CAPI 키(A):
<Try all certificates>
...
토큰 핀(N):
SSH 인증서 설정(S)
확인
취소

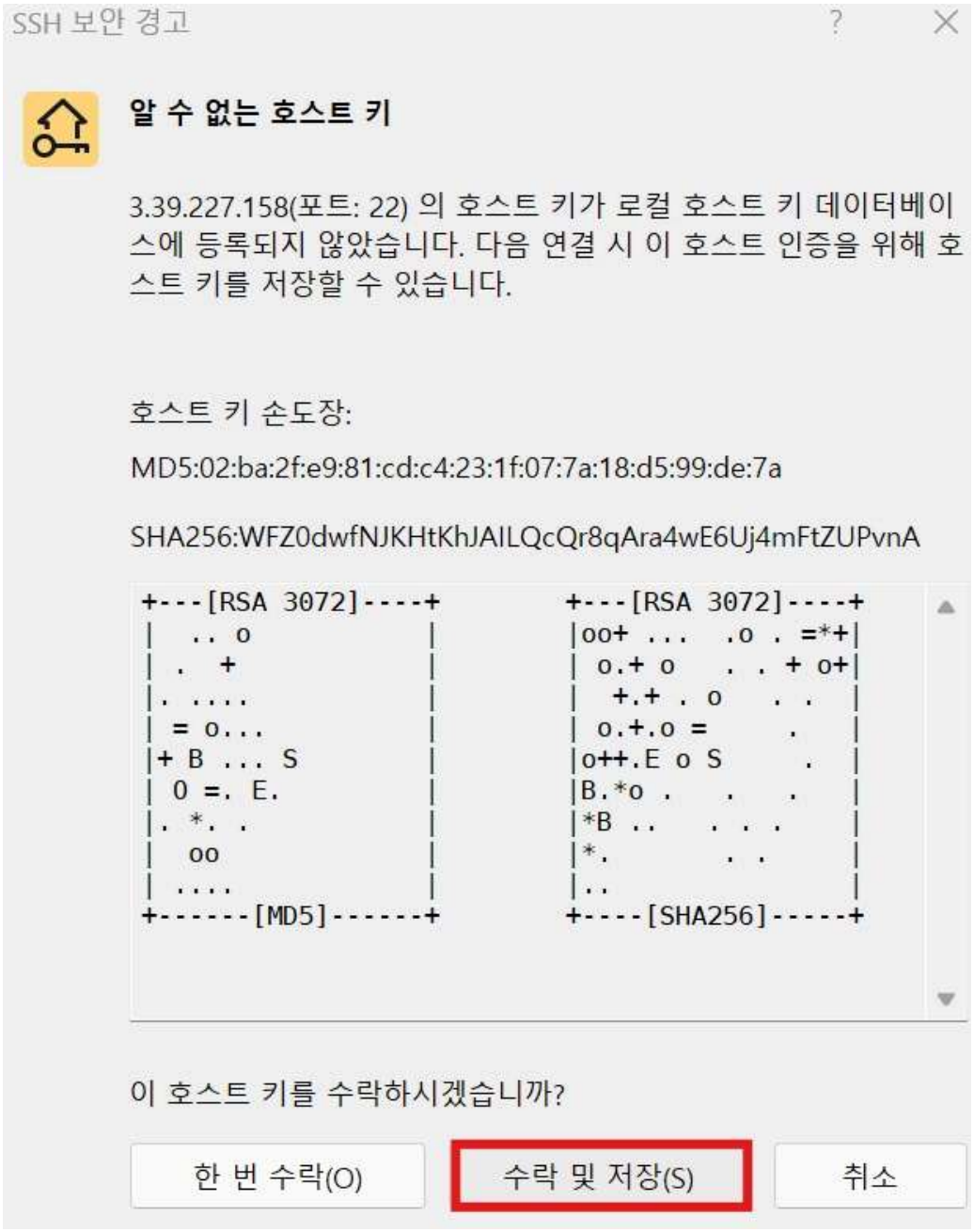

사용자 키
이름
종류
길이
설명
생성(G)...
등록 정보(P)
삭제(D)
가져오기(I)...
내보내기(E)...
확인
취소


열기
바탕 화면 › 키페어
키페어 검색
구성
새 폴더
갤러리
바탕 화면
다운로드
사진
음악
동영상
강사님 버전 이
이름
수정한 날짜
powermvp.pem
2025-05-08 오전 2:08
파일 이름(N): powermvp.pem
All Files (*.*)
열기(O)
취소

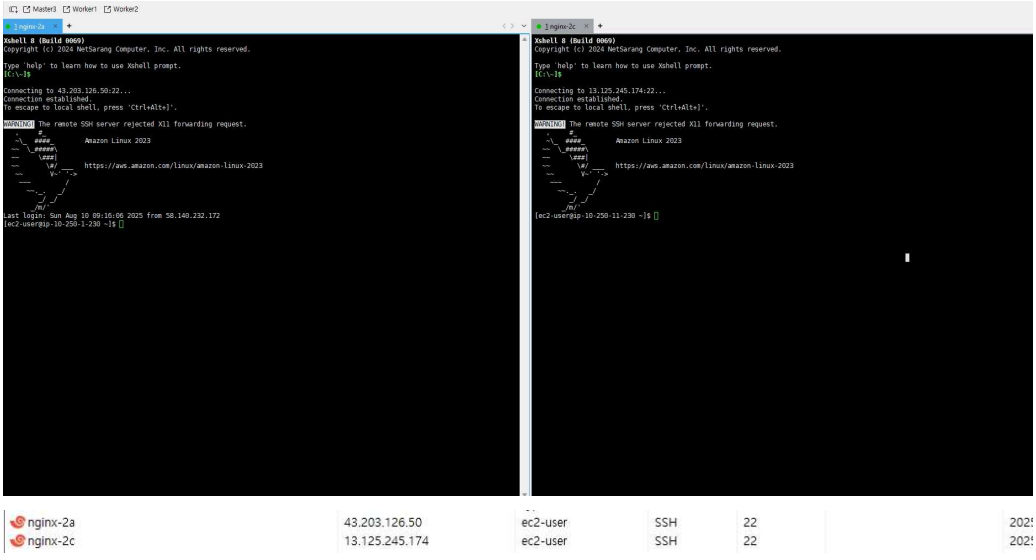




위와 같이 Public Key 가져오면 SSH 연결 시행



이와 같은 방식으로 ' VEC-PRD-VPC-NGINX-PUB-2A ', ' VEC-PRD-VPC-NGINX-PUB-2C ' 모두 등록



2) 만약 nginx web-server 가 Private Subnet에 존재한다면 Bastion-Server 를 통해 연결 후 작업 진행

```
ssh -i powermvp.pem ubuntu@10.250.1.240
```

```
ssh -i powermvp.pem ubuntu@10.250.11.240
```

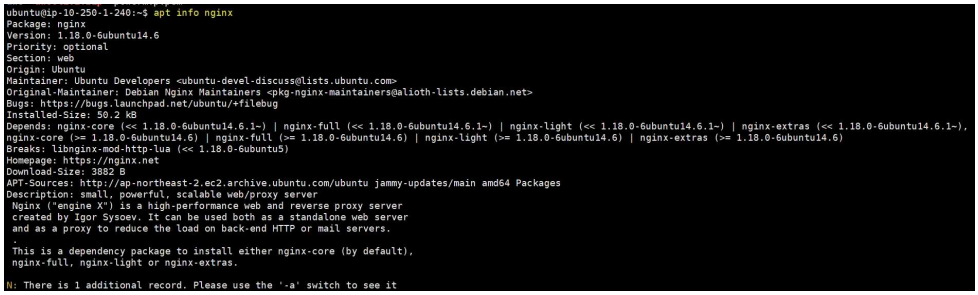
3) Nginx WEB-Server 설치 순서

[#0번작업] : Nginx 설치 가능한 버전 확인

`apt info nginx` 명령어로 name이 포함된 설치 가능한 패키지 목록을 조회한다.

```
sudo -i
```

```
apt info nginx
```



[#1번작업] : 저장소 파일 생성 및 내용 넣기

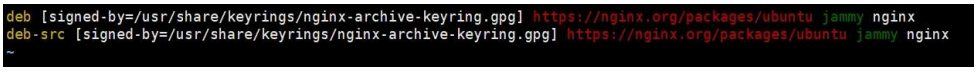
[root@ip-10-250-1-240]$ `sudo vi /etc/apt/sources.list.d/nginx.list`

```
sudo vi /etc/apt/sources.list.d/nginx.list
```

아래 코드 내용 복사하여 넣기

```
deb [signed-by=/usr/share/keyrings/nginx-archive-keyring.gpg]
```

```
deb-src [signed-by=/usr/share/keyrings/nginx-archive-keyring.
```



[root@ip-10-250-1-240]$ `curl -fsSL https://nginx.org/keys/nginx_signing.key | sudo gpg --dearmor -o /usr/share/keyrings/nginx-archive-keyring.gpg`

```
curl -fsSL https://nginx.org/keys/nginx_signing.key | sudo gp
```

Warning: apt-key is deprecated. Manage keyring files in trusted.gpg.d instead (see apt-key(8)).
OK

[#3번작업] : 저장소 업데이트 및 설치

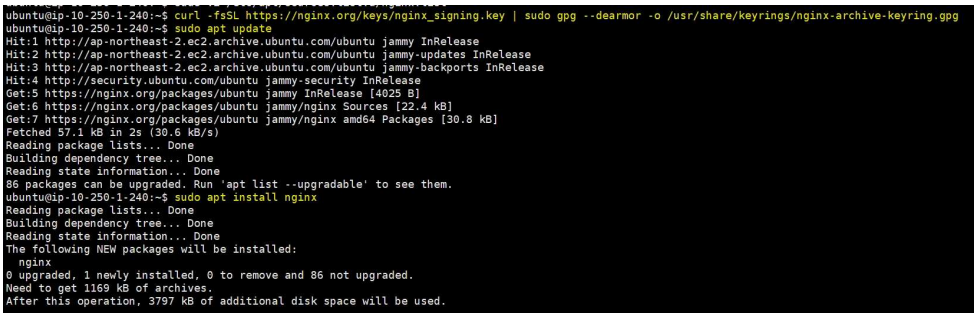
[root@ip-10-250-1-240]$ `sudo apt update`

```
sudo apt update
```

[#4번작업] : nginx 설치

[root@ip-10-250-1-240]$ `sudo apt install nginx`

```
sudo apt install nginx
```

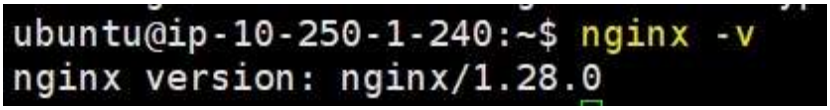


[#5번작업] : 설치 버전확인

[root@ip-10-250-1-240]$ **`nginx -v`**

```
nginx -v
```

nginx version: nginx/1.30.3



[#6번작업] : nginx 시작

[root@ip-10-250-11-240]$ `sudo systemctl start nginx`

```
sudo systemctl start nginx
```

[#7번작업] : nginx 상태 확인

[root@ip-10-250-11-240]$ `sudo systemctl status nginx`

```
sudo systemctl status nginx
```

● nginx.service - nginx - high performance web server
Loaded: loaded (/lib/systemd/system/nginx.service; enabled; vendor preset: enabled)
Active: active (running) since Sun 2025-07-06 11:54:28 UTC; 3s ago
Docs: https://nginx.org/en/docs/
Process: 2987 ExecStart=/usr/sbin/nginx -c ${CONFFILE} (code=exited, status=0/SUCCESS)
Main PID: 2988 (nginx)
Tasks: 2 (limit: 1129)
Memory: 3.4M
CPU: 10ms
CGroup: /system.slice/nginx.service
├─2988 "nginx: master process /usr/sbin/nginx -c /etc/nginx/nginx.conf"
└─2989 "nginx: worker process" "" "" "

```
ubuntu@ip-10-250-1-240:~$ sudo systemctl start nginx
ubuntu@ip-10-250-1-240:~$ sudo systemctl status nginx
● nginx.service - nginx - high performance web server
     Loaded: loaded (/lib/systemd/system/nginx.service; enabled; vendor preset: enabled)
     Active: active (running) since Thu 2025-08-07 15:50:47 UTC; 6s ago
       Docs: https://nginx.org/en/docs/
    Process: 2653 ExecStart=/usr/sbin/nginx -c ${CONFFILE} (code=exited, status=0/SUCCESS)
   Main PID: 2654 (nginx)
      Tasks: 3 (limit: 1077)
     Memory: 3.4M
        CPU: 10ms
     CGroup: /system.slice/nginx.service
             ├─2654 "nginx: master process /usr/sbin/nginx -c /etc/nginx/nginx.conf"
             ├─2655 "nginx: worker process" "" "" "" "" "" "" "" "" "" "" "" "" "" "" "" "" "" ""
             └─2656 "nginx: worker process" "" "" "" "" "" "" "" "" "" "" "" "" "" "" "" "" "" ""

Aug 07 15:50:47 ip-10-250-1-240 systemd[1]: Starting nginx - high performance web server...
Aug 07 15:50:47 ip-10-250-1-240 systemd[1]: Started nginx - high performance web server.
```

http://[Web서버 공인아이피] nginx 홈페이지가 뜨는지 확인한다.



==========

[여기서 잠깐]

이번에는 위에서 설치한 nginx-2a , nginx-2c 2대의 EC2를 ELB에 붙여보기 위해

[09]. ELB 설치_LoadBalancer 설치 로 이동하여 나머지를 구성후 다시 돌아 옵시다.

ELB 설치 구성하는 목적 3가지

첫째, 서버에 대한 부하분산

둘째, Health Check (건강한 서버에만 트래픽을 전달하겠다)

셋째, Autoscaling 구성 (서버 부하시 자동적인 서버증가를 통해 안정적인 트래픽 소화)

==========

03. nginx Web-server index.html 수정

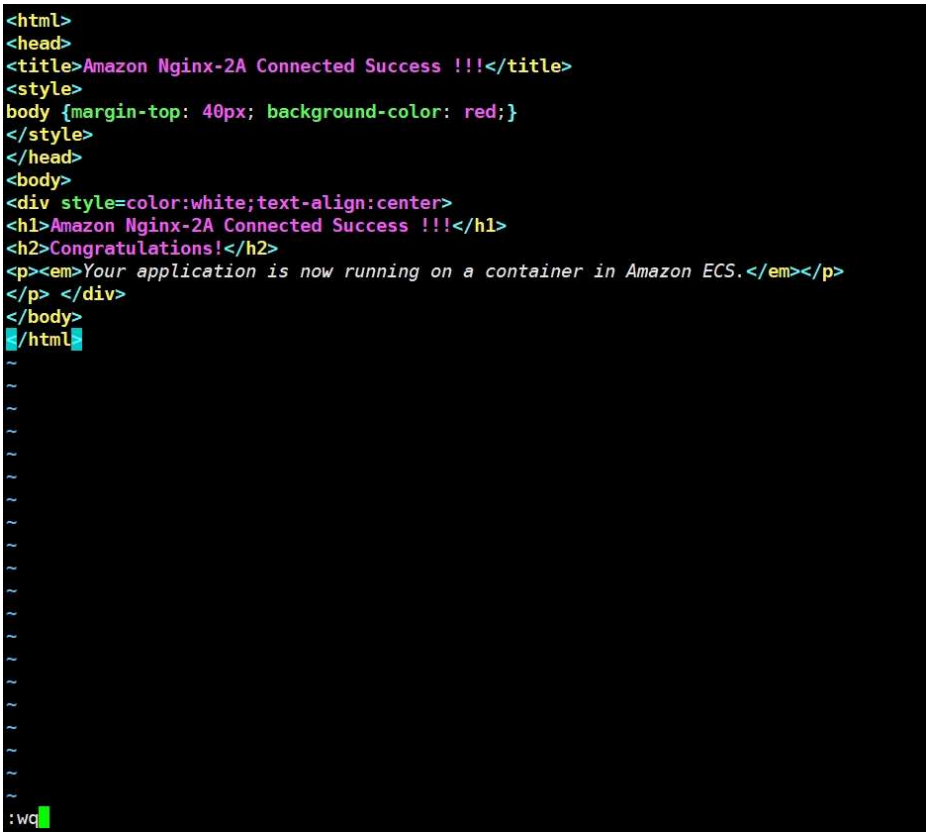
`VEC-PRD-VPC-NGINX-PUB-2A` 에서 진행 (root 로 이동하여 작업 진행)

```
sudo -i
```

: 아래 html 내용 복사하여 붙여넣고 `:wq` 로 저장

```
vi /usr/share/nginx/html/index.html
```

```
<html>
<head>
<title>Amazon Nginx-2A Connected Success !!!</title>
<style>
body {margin-top: 40px; background-color: red;}
</style>
</head>
<body>
<div style=color:white;text-align:center>
<h1>Amazon Nginx-2A Connected Success !!!</h1>
<h2>Congratulations!</h2>
<p><em>Your application is now running on a container in A
</p> </div>
</body>
</html>
```



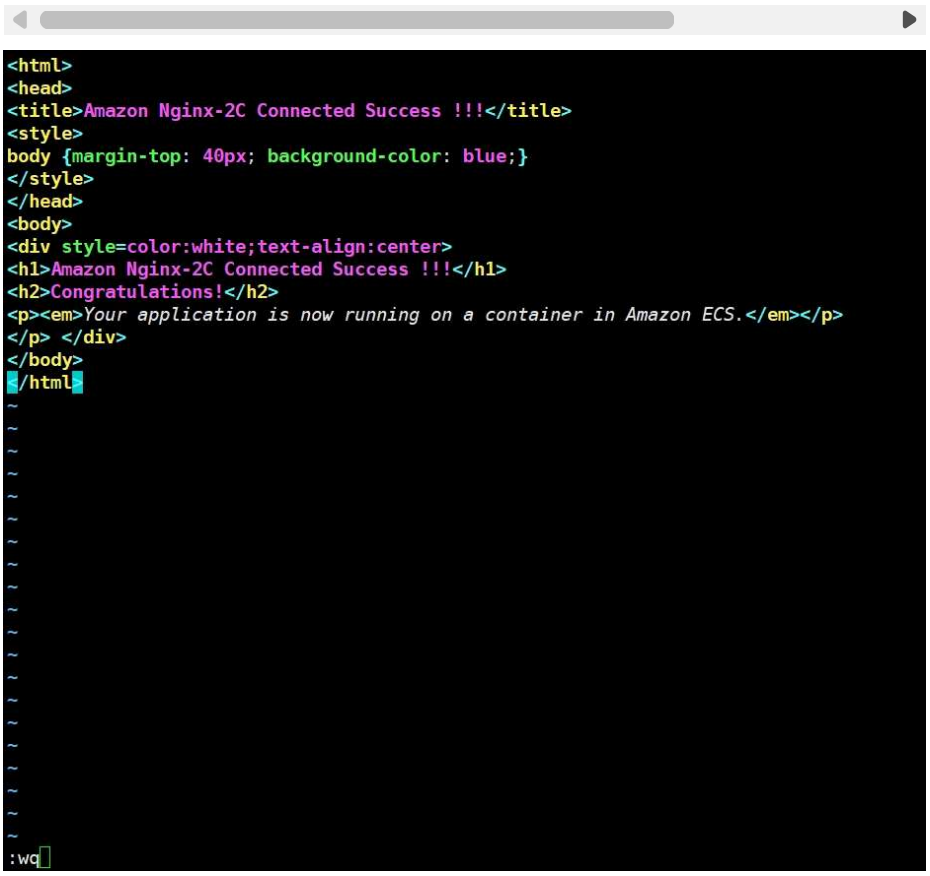
`VEC-PRD-VPC-NGINX-PUB-2C` 에서 진행 (root로 이동하여 작업 진행)

```
sudo -i
```

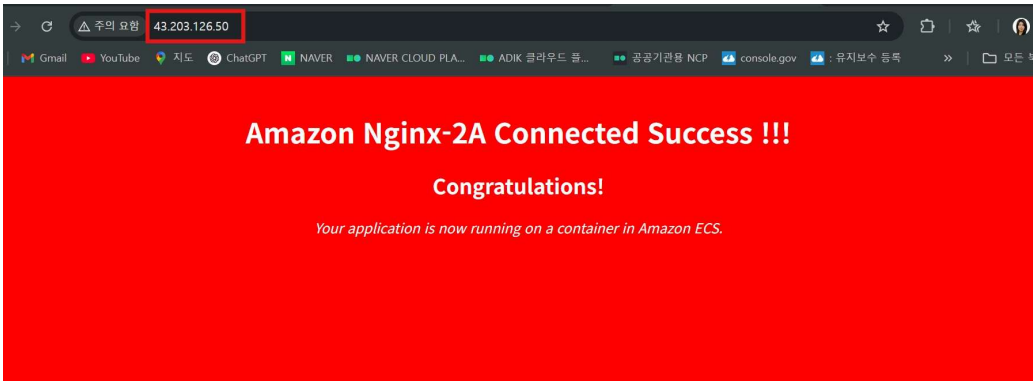
: 아래 html 내용 복사하여 붙여넣고 `[ESC] + :wq` 로 저장

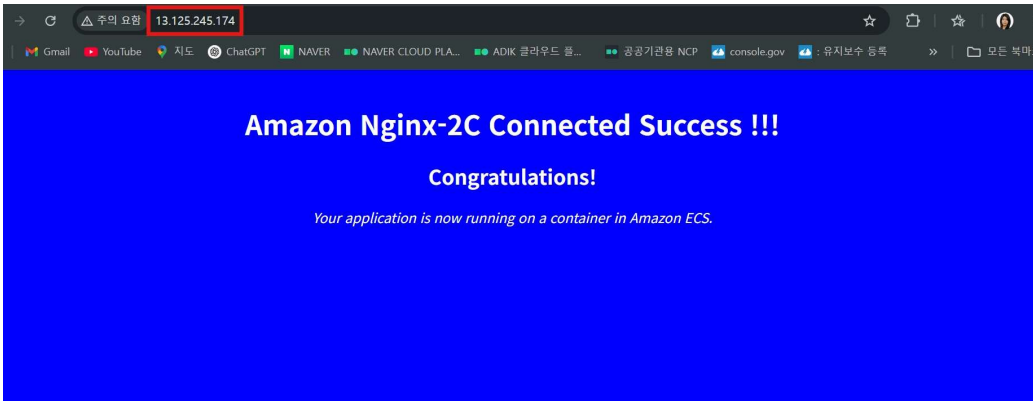
```
vi /usr/share/nginx/html/index.html
```

```
<html>
<head>
<title>Amazon Nginx-2C Connected Success !!!</title>
<style>
body {margin-top: 40px; background-color: blue;}
</style>
</head>
<body>
<div style=color:white;text-align:center>
<h1>Amazon Nginx-2C Connected Success !!!</h1>
<h2>Congratulations!</h2>
<p><em>Your application is now running on a container in A
</p> </div>
</body>
</html>
```



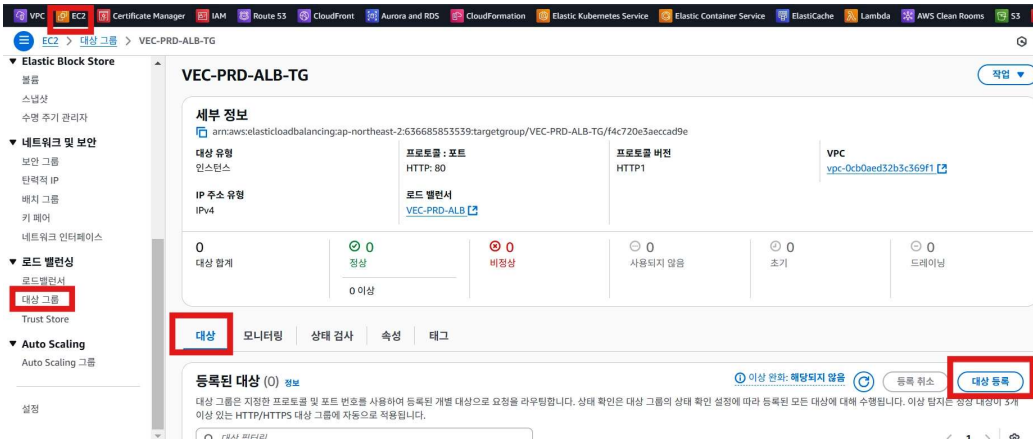
공인 IP로 접속 시 아래처럼 나옴





04. 새로 설치한 Nginx Web-Server 결과를 LoadBalancer로 확인하기

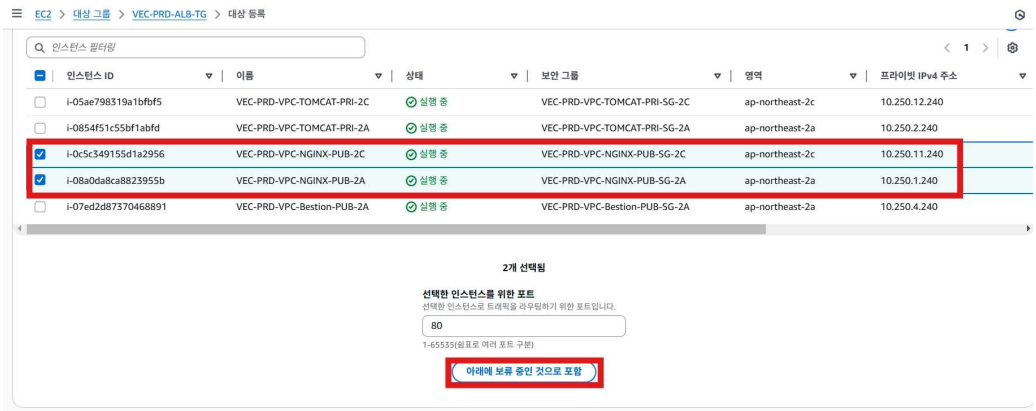
[EC2] - [대상그룹]에서 전날 생성한 ' VEC-PRD-ALB-TG ' 대상그룹 클릭하여 [대상] - [대상 등록] 클릭



[대상 등록]에서 이전에 생성한 EC2 선택 후 [아래에 보류 중인 것으로 포함] 클릭

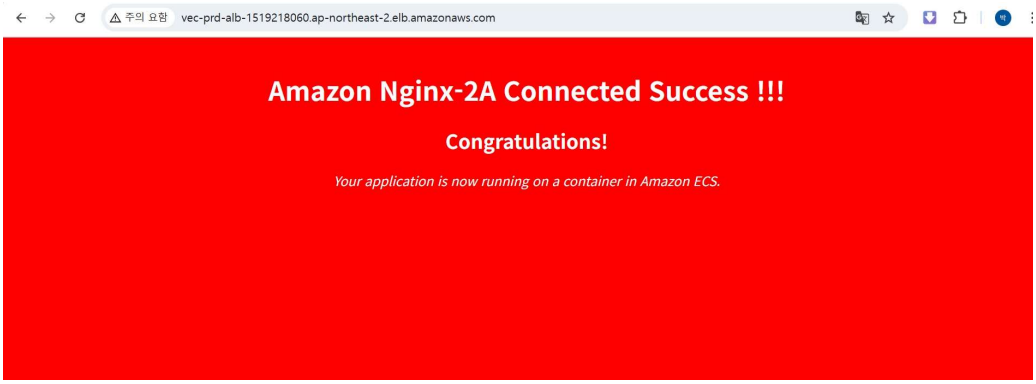
VEC-PRD-VPC-NGINX-PUB-2A

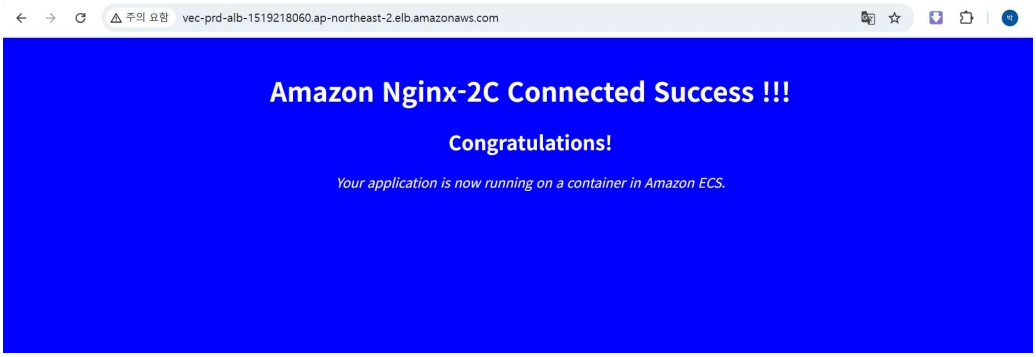
VEC-PRD-VPC-NGINX-PUB-2C



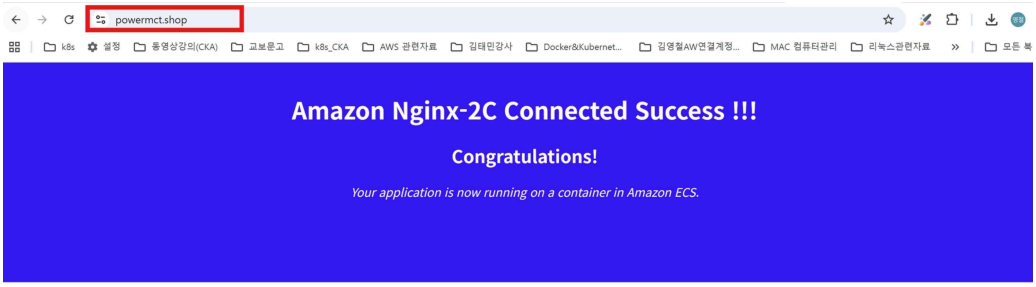
VEC-PRD-ALB DNS 접속시 아래와 같이 라운드로빈 방식으로 Nginx-2A , Nginx-2C 연결이 된다.

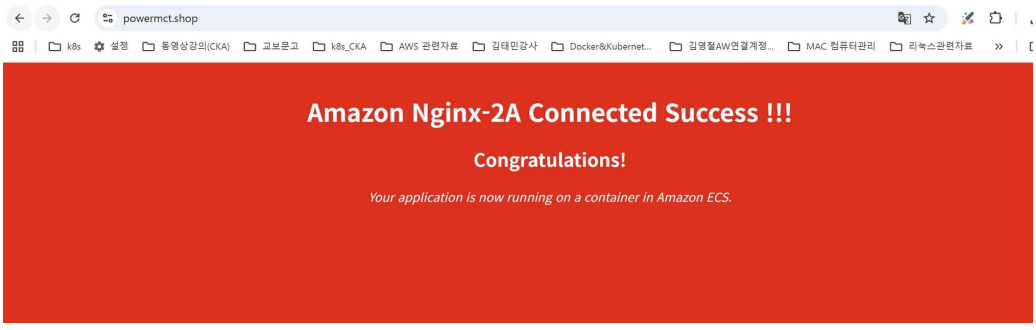
ALB 도메인으로 접속하면 정상적으로 nginx가 설치된 것을 확인할 수 있음





각자 Route53 등록되어 있는 도메인으로도 화면이 잘 뜬다.





05. Bastion-Server 접속후 TOMCAT 설치 (Tomcat2a, Tomcat2c 모두 설치)

[#0번작업] : JDK, Tomcat 설치 가능한 버전 확인
apt openjdk 명령어로 name이 포함된 설치 가능한 패키지 목록을 조회한다.

```
sudo -i
```

```
sudo -i
```

[#1번작업]
[root@ip-10-250-11-240]$ `apt search openjdk`

```
apt search openjdk
```

Sorting... Done
Full Text Search... Done
default-jdk/jammy 2:1.11-72build2 amd64
Standard Java or Java compatible Development Kit

default-jdk-doc/jammy 2:1.11-72build2 amd64
Standard Java or Java compatible Development Kit (documentation)

default-jdk-headless/jammy 2:1.11-72build2 amd64
Standard Java or Java compatible Development Kit (headless)

default-jre/jammy 2:1.11-72build2 amd64
Standard Java or Java compatible Runtime

default-jre-headless/jammy 2:1.11-72build2 amd64
Standard Java or Java compatible Runtime (headless)

openjdk-11-dbg/jammy-updates,jammy-security 11.0.27+6~us1-0ubuntu1~22.04 amd64
Java runtime based on OpenJDK (debugging symbols)

openjdk-11-doc/jammy-updates,jammy-security 11.0.27+6~us1-0ubuntu1~22.04 all
OpenJDK Development Kit (JDK) documentation

openjdk-11-jdk/jammy-updates,jammy-security 11.0.27+6~us1-0ubuntu1~22.04 amd64
OpenJDK Development Kit (JDK)

openjdk-11-jdk-headless/jammy-updates,jammy-security 11.0.27+6~us1-0ubuntu1~22.04 amd64
OpenJDK Development Kit (JDK) (headless)

openjdk-11-jre/jammy-updates,jammy-security 11.0.27+6~us1-0ubuntu1~22.04 amd64
OpenJDK Java runtime, using Hotspot JIT

openjdk-11-jre-headless/jammy-updates,jammy-security 11.0.27+6~us1-0ubuntu1~22.04 amd64
OpenJDK Java runtime, using Hotspot JIT (headless)

openjdk-11-source/jammy-updates,jammy-security 11.0.27+6~us1-0ubuntu1~22.04 all
OpenJDK Development Kit (JDK) source files

```
ubuntu@ip-10-250-2-240:~$ apt search openjdk
Sorting... Done
Full Text Search... Done
default-jdk/jammy 2:1.11-72build2 amd64
  Standard Java or Java compatible Development Kit

default-jdk-doc/jammy 2:1.11-72build2 amd64
  Standard Java or Java compatible Development Kit (documentation)

default-jdk-headless/jammy 2:1.11-72build2 amd64
  Standard Java or Java compatible Development Kit (headless)

default-jre/jammy 2:1.11-72build2 amd64
  Standard Java or Java compatible Runtime

default-jre-headless/jammy 2:1.11-72build2 amd64
  Standard Java or Java compatible Runtime (headless)

openjdk-11-dbg/jammy-updates,jammy-security 11.0.27+6~us1-0ubuntu1~22.04 amd64
```

[root@ip-10-250-11-240]$ `sudo apt update`

```
sudo apt update
```

```
ubuntu@ip-10-250-2-240:~$ sudo apt update
Hit:1 http://ap-northeast-2.ec2.archive.ubuntu.com/ubuntu jammy InRelease
Get:2 http://ap-northeast-2.ec2.archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Get:3 http://ap-northeast-2.ec2.archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]
Get:4 http://ap-northeast-2.ec2.archive.ubuntu.com/ubuntu jammy/universe amd64 Packages [14.1 MB]
Get:5 http://ap-northeast-2.ec2.archive.ubuntu.com/ubuntu jammy/universe Translation-en [5652 kB]
Get:6 http://ap-northeast-2.ec2.archive.ubuntu.com/ubuntu jammy/universe amd64 c-n-f Metadata [286 kB]
Get:7 http://ap-northeast-2.ec2.archive.ubuntu.com/ubuntu jammy/multiverse amd64 Packages [217 kB]
Get:8 http://ap-northeast-2.ec2.archive.ubuntu.com/ubuntu jammy/multiverse Translation-en [112 kB]
Get:9 http://ap-northeast-2.ec2.archive.ubuntu.com/ubuntu jammy/multiverse amd64 c-n-f Metadata [8372 B]
Get:10 http://ap-northeast-2.ec2.archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages [2802 kB]
Get:11 http://ap-northeast-2.ec2.archive.ubuntu.com/ubuntu jammy-updates/main Translation-en [443 kB]
Get:12 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
```

자바11 설치

아래 명령어를 입력하면 의존관계를 조회하고, 계속 설치를 진행할 것인지 묻는다.
y를 입력하면 의존관계를 다운로드하고 설치를 진행한다.

[#2번작업]

[root@ip-10-250-11-240]$ `sudo apt install openjdk-11-jdk -y`

```
sudo apt install openjdk-11-jdk -y
```



설치진행 화면 생략

[#3번작업] : 자바 버전 확인

java -version 명령어로 설치된 자바 버전을 확인한다.

[root@ip-10-250-11-240]$ `java -version`

```
java -version
```

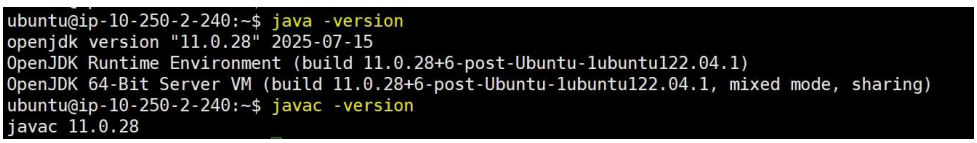
openjdk version "11.0.28" 2025-07-15
OpenJDK Runtime Environment (build 11.0.28+6-post-Ubuntu-1ubuntu122.04.1)
OpenJDK 64-Bit Server VM (build 11.0.28+6-post-Ubuntu-1ubuntu122.04.1, mixed mode, sharing)

[#4번작업] : 자바 컴포넌트 버전 확인

[root@ip-10-250-11-240]$ `javac -version`

```
javac -version
```

javac 11.0.28



==========**환경변수 설정**

(아래 #5번작업 ~ #10번작업까지 : 실행 안해줘도 된다)

[root@ip-10-250-11-240]$ `echo $JAVA_HOME`

```
echo $JAVA_HOME
```

[#5번작업]
설치 경로 확인
which 와 readlink 명령어를 통해 java와 javac가 설치된 경로를 확인한다.

아래명령어 실행
[root@ip-10-250-11-240]$ `which java`

```
which java
```

[ec2-user ~]$ `readlink -f /usr/bin/java`

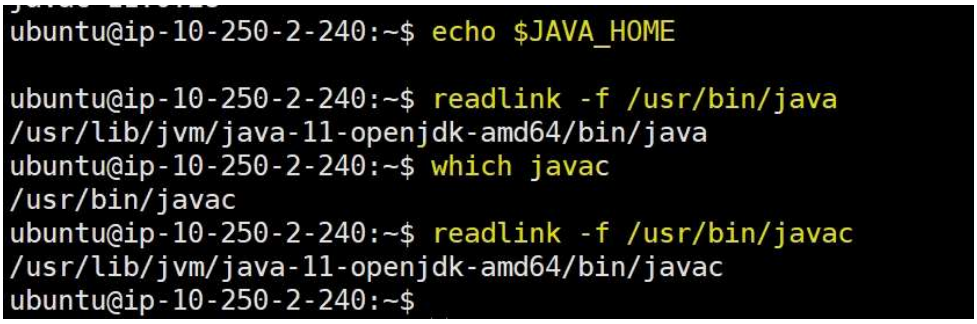
```
readlink -f /usr/bin/java
```

[#6번작업]
아래명령어 실행
[root@ip-10-250-11-240]$ `which javac`

```
which javac
```

[root@ip-10-250-11-240]$ `readlink -f /usr/bin/javac`

```
readlink -f /usr/bin/javac
```



환경변수 추가
export 명령어로 바로 환경변수에 추가할 수 있지만, 1회성으로 시스템 재시작 시 초기화된다.

[#7번작업]

[root@ip-10-250-11-240]$ `export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64`

```
export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64
```

[root@ip-10-250-11-240]$ `export PATH=$PATH:$JAVA_HOME/bin`

```
export PATH=$PATH:$JAVA_HOME/bin
```

[root@ip-10-250-11-240]$ `export CLASSPATH=$JAVA_HOME/lib/tools.jar`

```
export CLASSPATH=$JAVA_HOME/lib/tools.jar
```

echo 명령어로 등록한 환경변수 경로를 확인한다.

[#8번작업]

[root@ip-10-250-11-240]$ `echo $JAVA_HOME`

```
echo $JAVA_HOME
```

/usr/lib/jvm/java-11-openjdk-amd64

[#9번작업]

[root@ip-10-250-11-240]$ `echo $PATH`

```
echo $PATH
```

/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin:/usr/lib/jvm/java-11-openjdk-amd64/bin

[#10번작업]

[root@ip-10-250-11-240]$ `echo $CLASSPATH`

```
echo $CLASSPATH
```

/usr/lib/jvm/java-11-openjdk-amd64/lib/tools.jar

```
ubuntu@ip-10-250-2-240:~$ export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64
ubuntu@ip-10-250-2-240:~$ export PATH=$PATH:$JAVA_HOME/bin
ubuntu@ip-10-250-2-240:~$ export CLASSPATH=$JAVA_HOME/lib/tools.jar
ubuntu@ip-10-250-2-240:~$
ubuntu@ip-10-250-2-240:~$ echo $JAVA_HOME
/usr/lib/jvm/java-11-openjdk-amd64
ubuntu@ip-10-250-2-240:~$ echo $PATH
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin:/usr/lib/jvm/java-11-openjdk-amd64/
bin
ubuntu@ip-10-250-2-240:~$ echo $CLASSPATH
/usr/lib/jvm/java-11-openjdk-amd64/lib/tools.jar
```

==========

========== [#11번 작업부터 본격적으로 진행]

[#11번 작업]

[root@ip-10-250-11-240]$ `sudo vim /etc/profile`

```
sudo vim /etc/profile
```

```
export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64
export PATH=$PATH:$JAVA_HOME/bin
export CLASSPATH=$JAVA_HOME/lib/tools.jar
```

아래경로를 참고하여

/usr/lib/jvm/java-1.8.0-openjdk-1.8.0.432.b06-1.amzn2.0.1.x86_64

```
      . /etc/bash.bashrc
    fi
  else
    if [ "$(id -u)" -eq 0 ]; then
      PS1='# '
    else
      PS1='$ '
    fi
  fi
fi

if [ -d /etc/profile.d ]; then
  for i in /etc/profile.d/*.sh; do
    if [ -r $i ]; then
      . $i
    fi
  done
  unset i
fi
export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64
export PATH=$PATH:$JAVA_HOME/bin
export CLASSPATH=$JAVA_HOME/lib/tools.jar
```

(AWS) Ubuntu 24.04 버전에 Tomcat 9 설치하기

[#12번 작업]

[root@ip-10-250-11-240]$ `apt list | grep tomcat`

```
apt list | grep tomcat
```

```
ubuntu@ip-10-250-2-240:~$ apt list | grep tomcat

WARNING: apt does not have a stable CLI interface. Use with caution in scripts.

libtomcat9-embed-java/jammy-updates,jammy-security 9.0.58-1ubuntu0.2 all
libtomcat9-java/jammy-updates,jammy-security 9.0.58-1ubuntu0.2 all
libtomcatjss-java/jammy 8.0.0-2 all
tomcat-jakartaee-migration/jammy 0.2.0-1 all
tomcat9-admin/jammy-updates,jammy-security 9.0.58-1ubuntu0.2 all
tomcat9-common/jammy-updates,jammy-security 9.0.58-1ubuntu0.2 all
tomcat9-docs/jammy-updates,jammy-security 9.0.58-1ubuntu0.2 all
tomcat9-examples/jammy-updates,jammy-security 9.0.58-1ubuntu0.2 all
tomcat9-user/jammy-updates,jammy-security 9.0.58-1ubuntu0.2 all
tomcat9/jammy-updates,jammy-security 9.0.58-1ubuntu0.2 all
```

설치가능한 tomcat 패키지 목록
톰캣 웹사이트에서 9.0 버전 다운로드 주소 확인
톰캣 웹사이트에 가서 9.0 버전 파일의 주소를 확인하고 다운로드한다.

[#13번 작업] : 생략가능

아파치 톰캣 웹사이트 주소: https://tomcat.apache.org/download-90.cgi → 이미 해당 버전 다운로드 링크을 아래와 같이 표기 하였으므로 실행할 필요없다.

[#14번 작업]

```
ubuntu@ip-10-250-2-240:~$ wget https://archive.apache.org/dist/tomcat/tomcat-9/v9.0.106/bin/apache-tomcat-9.0.106.tar.gz
--2025-08-07 16:33:34--  https://archive.apache.org/dist/tomcat/tomcat-9/v9.0.106/bin/apache-tomcat-9.0.106.tar.gz
Resolving archive.apache.org (archive.apache.org)... 65.108.204.189, 2a01:4f9:1a:a084::2
Connecting to archive.apache.org (archive.apache.org)|65.108.204.189|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 13007554 (12M) [application/x-gzip]
Saving to: 'apache-tomcat-9.0.106.tar.gz'

apache-tomcat-9.0.106.tar.gz             100%[=====================================================================================>]  12.40M  3.59MB/s    in 3.5s

2025-08-07 16:33:39 (3.59 MB/s) - 'apache-tomcat-9.0.106.tar.gz' saved [13007554/13007554]
```

[주의 root에 작업한다]

root@ip-10-250-11-240]# `wget https://archive.apache.org/dist/tomcat/tomcat-9/v9.0.108/bin/apache-tomcat-9.0.108.tar.gz`

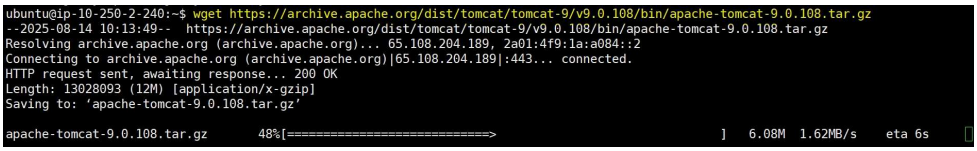
```
wget https://archive.apache.org/dist/tomcat/tomcat-9/v9.0.108
```

-2025-08-14 10:13:49-- https://archive.apache.org/dist/tomcat/tomcat-9/v9.0.108/bin/apache-tomcat-9.0.108.tar.gz
Resolving archive.apache.org (archive.apache.org)... 65.108.204.189, 2a01:4f9:1a:a084::2
Connecting to archive.apache.org (archive.apache.org)|65.108.204.189|:443... connected.
HTTP request sent, awaiting response... 200 OK

Length: 13028093 (12M) [application/x-gzip]
Saving to: ‘apache-tomcat-9.0.108.tar.gz’

apache-tomcat-9.0.108.tar.gz 100%
[==========>] 12.42M 3.36MB/s in 5.2s

2025-08-14 10:13:56 (2.40 MB/s) - ‘apache-tomcat-9.0.108.tar.gz’ saved [13028093/13028093]



100%
[==========>] 12,716,365 --.-K/s in 0.1s

2024-11-08 09:51:50 (89.2 MB/s) - ‘apache-tomcat-9.0.96.tar.gz’ saved [12716365/12716365]

[#15번 작업]

[root@ip-10-250-11-240]# `ls`

```
ls
```

apache-tomcat-9.0.102.tar.gz

[#16번 작업]

[root@ip-10-250-11-240]# `sudo groupadd tomcat`

```
sudo groupadd tomcat
```

[#16-1번 작업]

[root@ip-10-250-11-240]# `sudo mkdir /home/tomcat`

```
sudo mkdir /home/tomcat
```

[#16-2번 작업]

[root@ip-10-250-11-240]# `sudo tar -xf apache-tomcat-9.0.108.tar.gz -C /home/tomcat/`

```
sudo tar -xf apache-tomcat-9.0.108.tar.gz -C /home/tomcat/
```

[#16-3번 작업]

[root@ip-10-250-11-240]# `sudo useradd -s /usr/sbin/nologin -g tomcat -d /home/tomcat tomcat`

```
sudo useradd -s /usr/sbin/nologin -g tomcat -d /home/tomcat t
```

[#16-4번 작업]

root@ip-10-250-2-240:/# `ls /home/tomcat/apache-tomcat-9.0.108/`

```
ls /home/tomcat/apache-tomcat-9.0.108/
```

apache-tomcat-9.0.102/conf/
apache-tomcat-9.0.102/conf/catalina.policy
apache-tomcat-9.0.102/conf/catalina.properties
apache-tomcat-9.0.102/conf/context.xml
apache-tomcat-9.0.102/conf/jaspic-providers.xml
apache-tomcat-9.0.102/conf/jaspic-providers.xsd
apache-tomcat-9.0.102/conf/logging.properties
apache-tomcat-9.0.102/conf/server.xml
apache-tomcat-9.0.102/conf/tomcat-users.xml
apache-tomcat-9.0.102/conf/tomcat-users.xsd
apache-tomcat-9.0.102/conf/web.xml
apache-tomcat-9.0.102/bin/
apache-tomcat-9.0.102/lib/
apache-tomcat-9.0.102/logs/
apache-tomcat-9.0.102/temp/
apache-tomcat-9.0.102/webapps/
apache-tomcat-9.0.102/webapps/ROOT/
apache-tomcat-9.0.102/webapps/ROOT/WEB-INF/
apache-tomcat-9.0.102/webapps/docs/
apache-tomcat-9.0.102/webapps/docs/META-INF/

```
ubuntu@ip-10-250-2-240:~$ ls
apache-tomcat-9.0.108.tar.gz
ubuntu@ip-10-250-2-240:~$ sudo groupadd tomcat
ubuntu@ip-10-250-2-240:~$ sudo mkdir /home/tomcat
ubuntu@ip-10-250-2-240:~$ sudo tar -xf apache-tomcat-9.0.108.tar.gz -C /home/tomcat/
ubuntu@ip-10-250-2-240:~$ sudo useradd -s /usr/sbin/nologin -g tomcat -d /home/tomcat tomcat
ubuntu@ip-10-250-2-240:~$ ls /home/tomcat/apache-tomcat-9.0.108/
BUILDING.txt  CONTRIBUTING.md  LICENSE  NOTICE  README.md  RELEASE-NOTES  RUNNING.txt  bin  conf  lib  logs  temp  webapps  work
```

[#17번 작업]

[root@ip-10-250-11-240]# sudo vi /etc/systemd/system/tomcat.service

```
sudo vi /etc/systemd/system/tomcat.service
```

------복사해서 붙여놓는다.

```
[Unit]
Description=Apache Tomcat 9.0 Web Application Container
After=network.target

[Service]
Type=forking
Environment=JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64
Environment=CATALINA_PID=/home/tomcat/apache-tomcat-9.0.108/t
Environment=CATALINA_HOME=/home/tomcat/apache-tomcat-9.0.108
Environment=CATALINA_BASE=/home/tomcat/apache-tomcat-9.0.108
Environment='CATALINA_OPTS=-Xms512M -Xmx1024M -server -XX:+Us
Environment='JAVA_OPTS=-Djava.awt.headless=true -Djava.securi
ExecStart=/home/tomcat/apache-tomcat-9.0.108/bin/startup.sh
ExecStop=/home/tomcat/apache-tomcat-9.0.108/bin/shutdown.sh
User=tomcat
Group=tomcat
UMask=0007
RestartSec=10
Restart=always

[Install]

WantedBy=multi-user.target
```

---복사해서 붙여놓는다.

```
[Unit]
Description=Apache Tomcat 9.0 Web Application Container
After=network.target

[Service]
Type=forking
Environment=JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64
Environment=CATALINA_PID=/home/tomcat/apache-tomcat-9.0.108/temp/tomcat.pid
Environment=CATALINA_HOME=/home/tomcat/apache-tomcat-9.0.108
Environment=CATALINA_BASE=/home/tomcat/apache-tomcat-9.0.108
Environment='CATALINA_OPTS=-Xms512M -Xmx1024M -server -XX:+UseParallelGC'
Environment='JAVA_OPTS=-Djava.awt.headless=true -Djava.security.egd=file:/dev/./urandom'
ExecStart=/home/tomcat/apache-tomcat-9.0.108/bin/startup.sh
ExecStop=/home/tomcat/apache-tomcat-9.0.108/bin/shutdown.sh
User=tomcat
Group=tomcat
UMask=0007
RestartSec=10
Restart=always

[Install]
WantedBy=multi-user.target
```

[#18번 작업]

[root@ip-10-250-11-240]# `sudo systemctl daemon-reload`

```
sudo systemctl daemon-reload
```

[#19번 작업]

[root@ip-10-250-11-240]# `sudo systemctl enable tomcat.service`

```
sudo systemctl enable tomcat.service
```

[#20번 작업]

[root@ip-10-250-11-240]# `sudo chgrp -R tomcat /home/tomcat/`

```
sudo chgrp -R tomcat /home/tomcat/
```

[#21번 작업]

[root@ip-10-250-11-240]# `sudo chown -R tomcat /home/tomcat/`

```
sudo chown -R tomcat /home/tomcat/
```

[#22번 작업]

[root@ip-10-250-11-240]# `sudo chmod +x /home/tomcat/apache-tomcat-9.0.108/bin/*.sh`

```
sudo chmod +x /home/tomcat/apache-tomcat-9.0.108/bin/*.sh
```

[#23번 작업]

[root@ip-10-250-11-240]# `sudo systemctl start tomcat`

```
sudo systemctl start tomcat
```

[#24번 작업]

[root@ip-10-250-11-240]# `ps -ef|grep tomcat`

```
ps -ef|grep tomcat
```

tomcat 9478 1 26 07:53 ? 00:00:02 /usr/lib/jvm/jre/bin/java -Djava.util.logging.config.file=/home/tomcat/apache-tomcat-9.0.100/conf/logging.properties -Djava.util.logging.manager=org.apache.juli.ClassLoaderLogManager -Djava.awt.headless=true -Djava.security.egd=file:/dev/./urandom -Djdk.tls.ephemeralDHKeySize=2048 -Djava.protocol.handler.pkgs=org.apache.catalina.webresources -Dsun.io.useCanonCaches=false -Dorg.apache.catalina.security.SecurityListener.UMASK=0027 -Xms256M -Xmx256M -server -XX:+UseParallelGC -Dignore.endorsed.dirs= -classpath /home/tomcat/apache-tomcat-9.0.100/bin/bootstrap.jar:/home/tomcat/apache-tomcat-9.0.100/bin/tomcat-juli.jar -Dcatalina.base=/home/tomcat/apachetomcat-9.0.100 -Dcatalina.home=/home/tomcat/apache-tomcat-9.0.100 -Djava.io.tmpdir=/home/tomcat/apache-tomcat-9.0.100/temp org.apache.catalina.startup.Bootstrap start
root 9871 6887 0 07:53 pts/0 00:00:00 grep --color=auto tomcat

```
root@ip-10-250-12-240:~# sudo groupadd tomcat
root@ip-10-250-12-240:~# sudo mkdir /home/tomcat
root@ip-10-250-12-240:~# sudo tar -xf apache-tomcat-9.0.108.tar.gz -C /home/tomcat/
root@ip-10-250-12-240:~# sudo useradd -s /usr/sbin/nologin -g tomcat -d /home/tomcat tomcat
root@ip-10-250-12-240:~# sudo vi /etc/systemd/system/tomcat.service
root@ip-10-250-12-240:~# sudo systemctl daemon-reload
root@ip-10-250-12-240:~# sudo systemctl enable tomcat.service
Created symlink /etc/systemd/system/multi-user.target.wants/tomcat.service → /etc/systemd/system/tomcat.service.
root@ip-10-250-12-240:~# sudo chgrp -R tomcat /home/tomcat/
root@ip-10-250-12-240:~# sudo chown -R tomcat /home/tomcat/
root@ip-10-250-12-240:~# sudo chmod +x /home/tomcat/apache-tomcat-9.0.108/bin/*.sh
root@ip-10-250-12-240:~# sudo systemctl start tomcat
root@ip-10-250-12-240:~# ps -ef|grep tomcat
tomcat      4863       1 94 10:25 ?        00:00:03 /usr/lib/jvm/java-11-openjdk-amd64/bin/java -Djava.util.logging.config.file=/home
tomcat/apache-tomcat-9.0.108/conf/logging.properties -Djava.util.logging.manager=org.apache.juli.ClassLoaderLogManager -Djava.awt.head
less=true -Djava.security.egd=file:/dev/./urandom -Djdk.tls.ephemeralDHKeySize=2048 -Djava.protocol.handler.pkgs=org.apache.catalina.w
ebresources -Dsun.io.useCanonCaches=false -Dorg.apache.catalina.security.SecurityListener.UMASK=0027 -Xms512M -Xmx1024M -server -XX:+U
seParallelGC -Dignore.endorsed.dirs= -classpath /home/tomcat/apache-tomcat-9.0.108/bin/bootstrap.jar:/home/tomcat/apache-tomcat-9.0.10
8/bin/tomcat-juli.jar -Dcatalina.base=/home/tomcat/apache-tomcat-9.0.108 -Dcatalina.home=/home/tomcat/apache-tomcat-9.0.108 -Djava.io.
tmpdir=/home/tomcat/apache-tomcat-9.0.108/temp org.apache.catalina.startup.Bootstrap start
root        4893    2063  0 10:25 pts/1    00:00:00 grep --color=auto tomcat
```

[#25번 작업] : Tomcat 서비스 영구적으로 시작하기

[root@ip-10-250-11-240] # `systemctl enable tomcat.service`

```
systemctl enable tomcat.service
```

Created symlink from /etc/systemd/system/multi-user.target.wants/tomcat.service to /etc/systemd/system/tomcat.service.

복사 클릭 버전 (참고용)

1. Root 계정 접속 → ssh 는 원격접속이므로 tomcat2a 먼저 진행 후 tomcat2c 진행!

```
ssh tomcat2a
```

```
ssh tomcat2c
```

```
sudo -i
```

2. 설치 가능한 JDK 버전 확인

```
apt search openjdk
```

```
ubuntu@ip-10-250-2-240:~$ apt search openjdk
Sorting... Done
Full Text Search... Done
default-jdk/jammy 2:1.11-72build2 amd64
  Standard Java or Java compatible Development Kit

default-jdk-doc/jammy 2:1.11-72build2 amd64
  Standard Java or Java compatible Development Kit (documentation)

default-jdk-headless/jammy 2:1.11-72build2 amd64
  Standard Java or Java compatible Development Kit (headless)

default-jre/jammy 2:1.11-72build2 amd64
  Standard Java or Java compatible Runtime

default-jre-headless/jammy 2:1.11-72build2 amd64
  Standard Java or Java compatible Runtime (headless)

openjdk-11-dbg/jammy-updates,jammy-security 11.0.27+6~us1-0ubuntu1~22.04 amd64
```

3. 패키지 업데이트

```
sudo apt update
```

```
ubuntu@ip-10-250-2-240:~$ sudo apt update
Hit:1 http://ap-northeast-2.ec2.archive.ubuntu.com/ubuntu jammy InRelease
Get:2 http://ap-northeast-2.ec2.archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Get:3 http://ap-northeast-2.ec2.archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]
Get:4 http://ap-northeast-2.ec2.archive.ubuntu.com/ubuntu jammy/universe amd64 Packages [14.1 MB]
Get:5 http://ap-northeast-2.ec2.archive.ubuntu.com/ubuntu jammy/universe Translation-en [5652 kB]
Get:6 http://ap-northeast-2.ec2.archive.ubuntu.com/ubuntu jammy/universe amd64 c-n-f Metadata [286 kB]
Get:7 http://ap-northeast-2.ec2.archive.ubuntu.com/ubuntu jammy/multiverse amd64 Packages [217 kB]
Get:8 http://ap-northeast-2.ec2.archive.ubuntu.com/ubuntu jammy/multiverse Translation-en [112 kB]
Get:9 http://ap-northeast-2.ec2.archive.ubuntu.com/ubuntu jammy/multiverse amd64 c-n-f Metadata [8372 B]
Get:10 http://ap-northeast-2.ec2.archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages [2802 kB]
Get:11 http://ap-northeast-2.ec2.archive.ubuntu.com/ubuntu jammy-updates/main Translation-en [443 kB]
Get:12 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
```

4. 자바 11 설치

```
sudo apt install openjdk-11-jdk -y
```

```
ubuntu@ip-10-250-2-240:~$ sudo apt install openjdk-11-jdk -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  alsa-topology-conf alsa-ucm-conf at-spi2-core ca-certificates-java dconf-gsettings-backend dconf-service fontconfig-config
  fonts-dejavu-core fonts-dejavu-extra gsettings-desktop-schemas java-common libasound2 libasound2-data libatk-bridge2.0-0
  libatk-wrapper-java libatk-wrapper-java-jni libatk1.0-0 libatk1.0-data libatspi2.0-0 libavahi-client3 libavahi-common-data
  libavahi-common3 libcups2 libdconf1 libdrm-amdgpu1 libdrm-intel1 libdrm-nouveau2 libdrm-radeon1 libfontconfig1 libfontenc1 libgif7
  libgl1 libgl1-amber-dri libgl1-mesa-dri libglapi-mesa libglvnd0 libglx-mesa0 libglx0 libgraphite2-3 libharfbuzz0b libice-dev
  libice6 libjpeg-turbo8 libjpeg8 liblcms2-2 libllvm15 libpciaccess0 libpcsclite1 libpthread-stubs0-dev libsensors-config
  libsensors5 libsm-dev libsm6 libx11-dev libx11-xcb1 libxau-dev libxaw7 libxcb-dri2-0 libxcb-dri3-0 libxcb-glx0 libxcb-present0
  libxcb-randr0 libxcb-shape0 libxcb-shm0 libxcb-sync1 libxcb-xfixes0 libxcb1-dev libxcomposite1 libxdmcp-dev libxfixes3 libxft2
  libxi6 libxinerama1 libxkbfile1 libxmu6 libxpm4 libxrandr2 libxrender1 libxshmfence1 libxt-dev libxt6 libxtst6 libxv1 libxxf86dga1
  libxxf86vm1 openjdk-11-jdk-headless openjdk-11-jre openjdk-11-jre-headless session-migration x11-common x11-utils x11proto-dev
  xorg-sgml-doctools xtrans-dev
Suggested packages:
```

5. 자바 버전 확인

```
export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64
```

```
javac -version
```

```
ubuntu@ip-10-250-2-240:~$ java -version
openjdk version "11.0.28" 2025-07-15
OpenJDK Runtime Environment (build 11.0.28+6-post-Ubuntu-1ubuntu122.04.1)
OpenJDK 64-Bit Server VM (build 11.0.28+6-post-Ubuntu-1ubuntu122.04.1, mixed mode, sharing)
ubuntu@ip-10-250-2-240:~$ javac -version
javac 11.0.28
```

==========

==========

5. 환경변수 설정

```
echo $JAVA_HOME
```

```
which java
```

```
readlink -f /usr/bin/java
```

```
which javac
```

```
readlink -f /usr/bin/javac
```

```
ubuntu@ip-10-250-2-240:~$ echo $JAVA_HOME

ubuntu@ip-10-250-2-240:~$ readlink -f /usr/bin/java
/usr/lib/jvm/java-11-openjdk-amd64/bin/java
ubuntu@ip-10-250-2-240:~$ which javac
/usr/bin/javac
ubuntu@ip-10-250-2-240:~$ readlink -f /usr/bin/javac
/usr/lib/jvm/java-11-openjdk-amd64/bin/javac
ubuntu@ip-10-250-2-240:~$
```

6. 환경변수 설정 및 확인

```
export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64
```

```
export PATH=$PATH:$JAVA_HOME/bin
```

```
export CLASSPATH=$JAVA_HOME/lib/tools.jar
```

```
echo $JAVA_HOME
```

```
echo $PATH
```

```
echo $CLASSPATH
```

```
ubuntu@ip-10-250-2-240:~$ export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64
ubuntu@ip-10-250-2-240:~$ export PATH=$PATH:$JAVA_HOME/bin
ubuntu@ip-10-250-2-240:~$ export CLASSPATH=$JAVA_HOME/lib/tools.jar
ubuntu@ip-10-250-2-240:~$
ubuntu@ip-10-250-2-240:~$ echo $JAVA_HOME
/usr/lib/jvm/java-11-openjdk-amd64
ubuntu@ip-10-250-2-240:~$ echo $PATH
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin:/usr/lib/jvm/java-11-openjdk-amd64/
bin
ubuntu@ip-10-250-2-240:~$ echo $CLASSPATH
/usr/lib/jvm/java-11-openjdk-amd64/lib/tools.jar
```

==========

==========

7. 재기동 시 환경변수 적용하도록 설정

```
sudo vim /etc/profile
```

```
export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64
export PATH=$PATH:$JAVA_HOME/bin
export CLASSPATH=$JAVA_HOME/lib/tools.jar
```

```
      . /etc/bash.bashrc
    fi
  else
    if [ "$(id -u)" -eq 0 ]; then
      PS1='# '
    else
      PS1='$ '
    fi
  fi
fi

if [ -d /etc/profile.d ]; then
  for i in /etc/profile.d/*.sh; do
    if [ -r $i ]; then
      . $i
    fi
  done
  unset i
fi
export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64
export PATH=$PATH:$JAVA_HOME/bin
export CLASSPATH=$JAVA_HOME/lib/tools.jar
```

---

8. 설치 가능한 톰캣 버전 확인하기

```
apt list | grep tomcat
```

```
ubuntu@ip-10-250-2-240:~$ apt list | grep tomcat

WARNING: apt does not have a stable CLI interface. Use with caution in scripts.

libtomcat9-embed-java/jammy-updates,jammy-security 9.0.58-1ubuntu0.2 all
libtomcat9-java/jammy-updates,jammy-security 9.0.58-1ubuntu0.2 all
libtomcatjss-java/jammy 8.0.0-2 all
tomcat-jakartaee-migration/jammy 0.2.0-1 all
tomcat9-admin/jammy-updates,jammy-security 9.0.58-1ubuntu0.2 all
tomcat9-common/jammy-updates,jammy-security 9.0.58-1ubuntu0.2 all
tomcat9-docs/jammy-updates,jammy-security 9.0.58-1ubuntu0.2 all
tomcat9-examples/jammy-updates,jammy-security 9.0.58-1ubuntu0.2 all
tomcat9-user/jammy-updates,jammy-security 9.0.58-1ubuntu0.2 all
tomcat9/jammy-updates,jammy-security 9.0.58-1ubuntu0.2 all
```

9. 톰캣 설치 파일 다운로드

```
wget https://archive.apache.org/dist/tomcat/tomcat-9/v9.0.
```

```
ubuntu@ip-10-250-2-240:~$ wget https://archive.apache.org/dist/tomcat/tomcat-9/v9.0.108/bin/apache-tomcat-9.0.108.tar.gz
--2025-08-14 10:13:49--  https://archive.apache.org/dist/tomcat/tomcat-9/v9.0.108/bin/apache-tomcat-9.0.108.tar.gz
Resolving archive.apache.org (archive.apache.org)... 65.108.204.189, 2a01:4f9:1a:a084::2
Connecting to archive.apache.org (archive.apache.org)|65.108.204.189|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 13028093 (12M) [application/x-gzip]
Saving to: 'apache-tomcat-9.0.108.tar.gz'

apache-tomcat-9.0.108.tar.gz       48%[=========================>                       ]   6.08M  1.62MB/s    eta 6s
```

10. tomcat 계정, 그룹 생성 및 설치파일 압축 풀기

```
sudo groupadd tomcat
```

```
sudo mkdir /home/tomcat
```

```
sudo tar -xf apache-tomcat-9.0.108.tar.gz -C /home/tomcat/
```

```
sudo useradd -s /usr/sbin/nologin -g tomcat -d /home/tomca
```

```
ubuntu@ip-10-250-2-240:~$ ls
apache-tomcat-9.0.108.tar.gz
ubuntu@ip-10-250-2-240:~$ sudo groupadd tomcat
ubuntu@ip-10-250-2-240:~$ sudo mkdir /home/tomcat
ubuntu@ip-10-250-2-240:~$ sudo tar -xf apache-tomcat-9.0.108.tar.gz -C /home/tomcat/
ubuntu@ip-10-250-2-240:~$ sudo useradd -s /usr/sbin/nologin -g tomcat -d /home/tomcat tomcat
ubuntu@ip-10-250-2-240:~$ ls /home/tomcat/apache-tomcat-9.0.108/
BUILDING.txt  CONTRIBUTING.md  LICENSE  NOTICE  README.md  RELEASE-NOTES  RUNNING.txt  bin  conf  lib  logs  temp  webapps  work
```

11. 톰캣 설정파일 수정

```
sudo vi /etc/systemd/system/tomcat.service
```

----------복사해서 붙여놓는다.

```
[Unit]
Description=Apache Tomcat 9.0 Web Application Container
After=network.target

[Service]
Type=forking
Environment=JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64
Environment=CATALINA_PID=/home/tomcat/apache-tomcat-9.0.10
Environment=CATALINA_HOME=/home/tomcat/apache-tomcat-9.0.1
Environment=CATALINA_BASE=/home/tomcat/apache-tomcat-9.0.1
Environment='CATALINA_OPTS=-Xms512M -Xmx1024M -server -XX:
Environment='JAVA_OPTS=-Djava.awt.headless=true -Djava.sec
ExecStart=/home/tomcat/apache-tomcat-9.0.108/bin/startup.s
ExecStop=/home/tomcat/apache-tomcat-9.0.108/bin/shutdown.s
User=tomcat
Group=tomcat
UMask=0007
RestartSec=10
Restart=always

[Install]

WantedBy=multi-user.target
```

------복사해서 붙여놓는다.

```
[Unit]
Description=Apache Tomcat 9.0 Web Application Container
After=network.target

[Service]
Type=forking
Environment=JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64
Environment=CATALINA_PID=/home/tomcat/apache-tomcat-9.0.108/temp/tomcat.pid
Environment=CATALINA_HOME=/home/tomcat/apache-tomcat-9.0.108
Environment=CATALINA_BASE=/home/tomcat/apache-tomcat-9.0.108
Environment='CATALINA_OPTS=-Xms512M -Xmx1024M -server -XX:+UseParallelGC'
Environment='JAVA_OPTS=-Djava.awt.headless=true -Djava.security.egd=file:/dev/./urandom'
ExecStart=/home/tomcat/apache-tomcat-9.0.108/bin/startup.sh
ExecStop=/home/tomcat/apache-tomcat-9.0.108/bin/shutdown.sh
User=tomcat
Group=tomcat
UMask=0007
RestartSec=10
Restart=always

[Install]
WantedBy=multi-user.target
```

12. 설정파일 리로드 및 톰캣 실행

```
sudo systemctl daemon-reload
```

```
sudo systemctl enable tomcat.service
```

```
sudo chgrp -R tomcat /home/tomcat/
```

```
sudo chown -R tomcat /home/tomcat/
```

```
sudo chmod +x /home/tomcat/apache-tomcat-9.0.108/bin/*.sh
```

```
sudo systemctl start tomcat
```

```
ps -ef|grep tomcat
```

```
root@ip-10-250-12-240:~# sudo groupadd tomcat
root@ip-10-250-12-240:~# sudo mkdir /home/tomcat
root@ip-10-250-12-240:~# sudo tar -xf apache-tomcat-9.0.108.tar.gz -C /home/tomcat/
root@ip-10-250-12-240:~# sudo useradd -s /usr/sbin/nologin -g tomcat -d /home/tomcat tomcat
root@ip-10-250-12-240:~# sudo vi /etc/systemd/system/tomcat.service
root@ip-10-250-12-240:~# sudo systemctl daemon-reload
root@ip-10-250-12-240:~# sudo systemctl enable tomcat.service
Created symlink /etc/systemd/system/multi-user.target.wants/tomcat.service → /etc/systemd/system/tomcat.service.
root@ip-10-250-12-240:~# sudo chgrp -R tomcat /home/tomcat/
root@ip-10-250-12-240:~# sudo chown -R tomcat /home/tomcat/
root@ip-10-250-12-240:~# sudo chmod +x /home/tomcat/apache-tomcat-9.0.108/bin/*.sh
root@ip-10-250-12-240:~# sudo systemctl start tomcat
root@ip-10-250-12-240:~# ps -ef|grep tomcat
tomcat      4863       1 94 10:25 ?        00:00:03 /usr/lib/jvm/java-11-openjdk-amd64/bin/java -Djava.util.logging.config.file=/home
tomcat/apache-tomcat-9.0.108/conf/logging.properties -Djava.util.logging.manager=org.apache.juli.ClassLoaderLogManager -Djava.awt.head
less=true -Djava.security.egd=file:/dev/./urandom -Djdk.tls.ephemeralDHKeySize=2048 -Djava.protocol.handler.pkgs=org.apache.catalina.w
ebresources -Dsun.io.useCanonCaches=false -Dorg.apache.catalina.security.SecurityListener.UMASK=0027 -Xms512M -Xmx1024M -server -XX:+U
seParallelGC -Dignore.endorsed.dirs= -classpath /home/tomcat/apache-tomcat-9.0.108/bin/bootstrap.jar:/home/tomcat/apache-tomcat-9.0.10
8/bin/tomcat-juli.jar -Dcatalina.base=/home/tomcat/apache-tomcat-9.0.108 -Dcatalina.home=/home/tomcat/apache-tomcat-9.0.108 -Djava.io.
tmpdir=/home/tomcat/apache-tomcat-9.0.108/temp org.apache.catalina.startup.Bootstrap start
root        4893    2063  0 10:25 pts/1    00:00:00 grep --color=auto tomcat
```

13. 톰캣 서비스 확인하기

```
systemctl enable tomcat.service
```

06. nginx와 tomca간의 통신 설정하기 (nginx2a, nginx2c 모두에서 진행)

nginx web 서버와 tomcat was 서버 연결 (nginx2a 에서 작업 진행)

#1번작업 : nginx 설정 파일 수정

[root@ip-10-250-1-240]$

```
sudo vi /etc/nginx/conf.d/default.conf
```

```
sudo vi /etc/nginx/conf.d/default.conf
```

아래 코드 내용 복사하여 넣기 → tomcat2a와 연결

```
server {
        listen 80;
        server_name localhost;

        # 톰캣 서버(10.250.2.240:8080)로 모든 요청 프록시
        location / {
            proxy_pass http://10.250.2.240:8080;
            proxy_set_header Host $host;
            proxy_set_header X-Real-IP $remote_addr;
            proxy_set_header X-Forwarded-For $proxy_add_x_
            proxy_set_header X-Forwarded-Proto $scheme;
            proxy_connect_timeout 300;
            proxy_send_timeout 300;
            proxy_read_timeout 300;
        }
    }
```

```
error_log  /var/log/nginx/error.log notice;
pid        /run/nginx.pid;


events {
    worker_connections  1024;
}


http {
    server {
        listen 80;
        server_name localhost;

        # 톰캣 서버(10.2.250.240:8080)로 모든 요청 프록시
        location / {
            proxy_pass http://10.250.2.240:8080;
            proxy_set_header Host $host;
            proxy_set_header X-Real-IP $remote_addr;
            proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
            proxy_set_header X-Forwarded-Proto $scheme;
            proxy_connect_timeout 300;
            proxy_send_timeout 300;
            proxy_read_timeout 300;
        }
    }
```

#2번작업 - GPG 키 추가

[root@ip-10-250-1-240]$ `sudo nginx -t`

```
sudo nginx -t
```

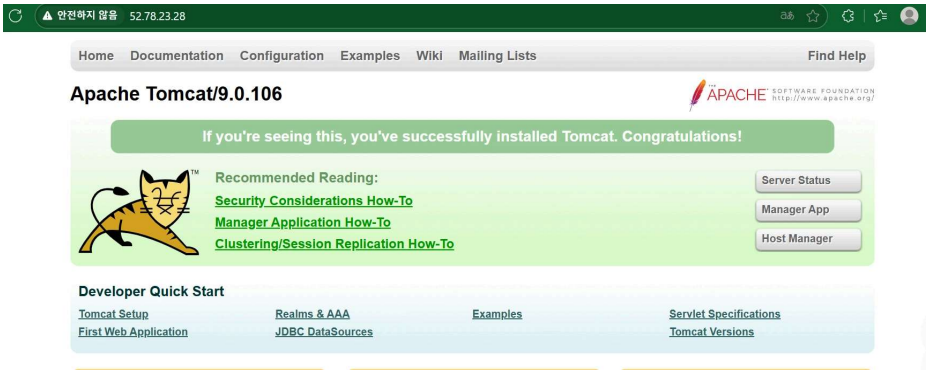
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful

#3번작업 - nginx 재가동

[root@ip-10-250-1-240]$ `sudo systemctl restart nginx`

```
sudo systemctl restart nginx
```

1. 재가동 후 nginx-2A 공인IP 로 접속하면 tomcat 야옹이 화면을 볼 수 있음



nginx web 서버와 tomcat was 서버 연결 (nginx2c 에서 작업)

#1번작업 - nginx 설정 파일 수정

[root@ip-10-250-11-240]$

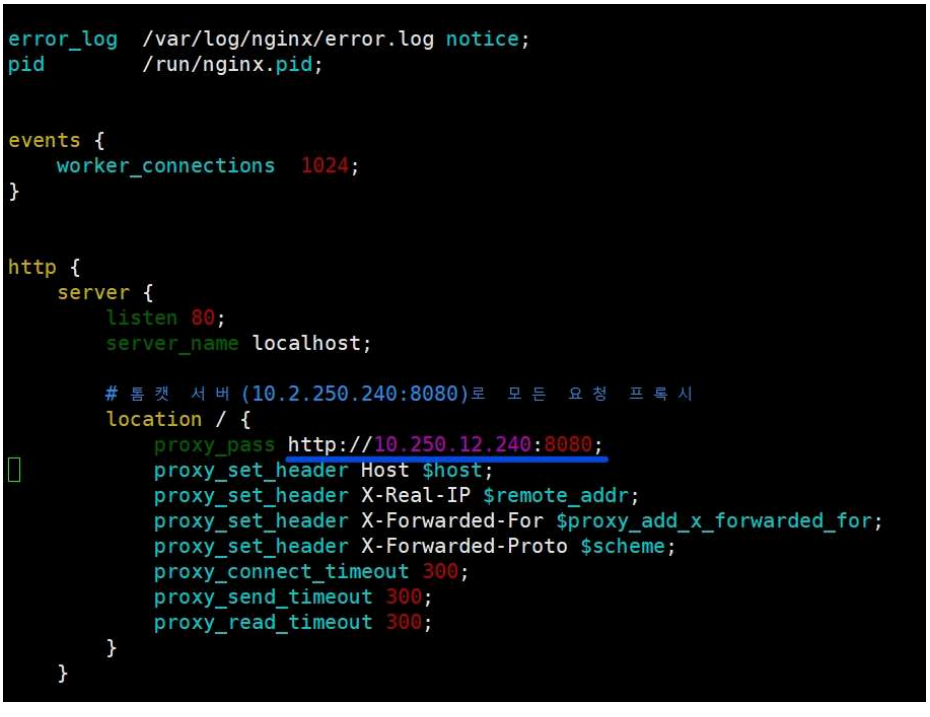
```
sudo vi /etc/nginx/conf.d/default.conf
```

```
sudo vi /etc/nginx/conf.d/default.conf
```

아래 코드 내용 복사하여 붙여넣기 → tomcat2c와 연결

```
server {
        listen 80;
        server_name localhost;

        # 톰캣 서버(10.250.12.240:8080)로 모든 요청 프록시
        location / {
            proxy_pass http://10.250.12.240:8080;
            proxy_set_header Host $host;
            proxy_set_header X-Real-IP $remote_addr;
            proxy_set_header X-Forwarded-For $proxy_add_x_
            proxy_set_header X-Forwarded-Proto $scheme;
            proxy_connect_timeout 300;
            proxy_send_timeout 300;
            proxy_read_timeout 300;
        }
    }
```



#2번작업 - GPG 키 추가

[root@ip-10-250-1-240]$ `sudo nginx -t`

```
sudo nginx -t
```

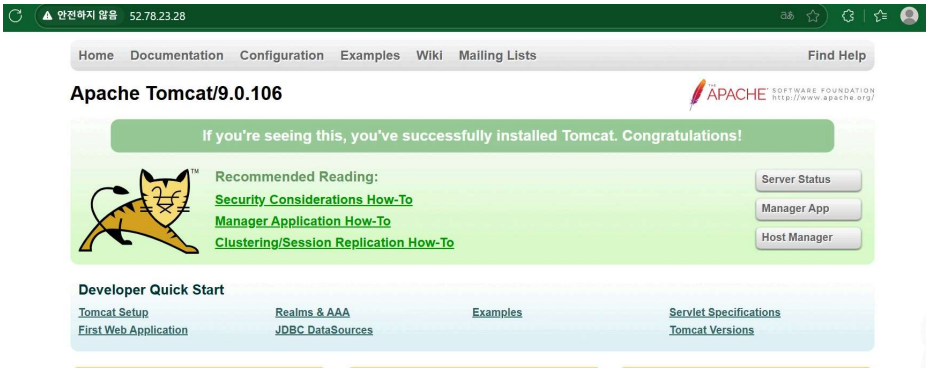
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful

#3번작업 - nginx 재가동

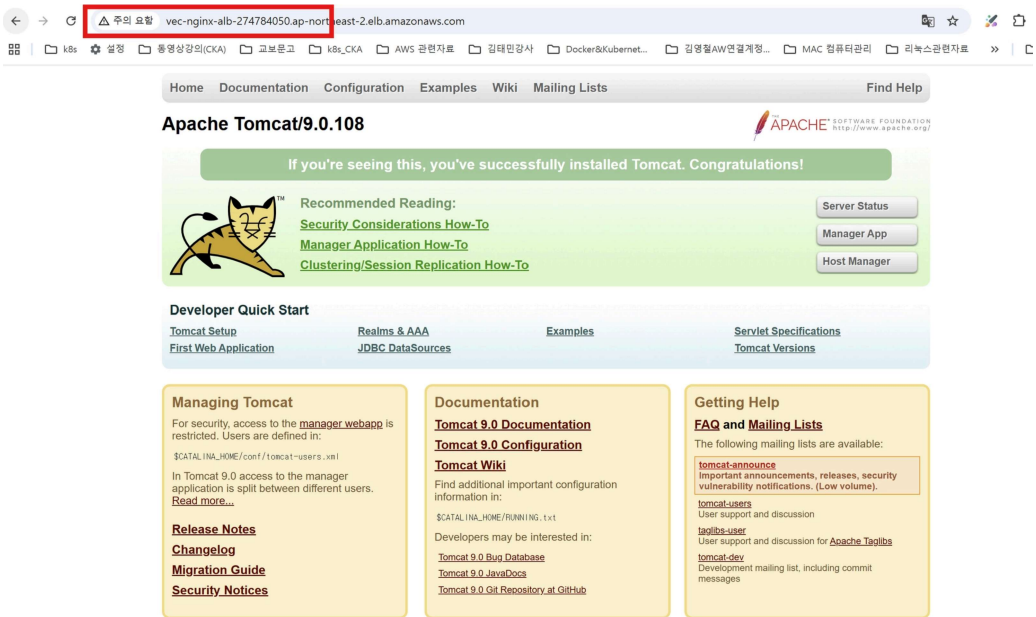
[root@ip-10-250-1-240]$ `sudo systemctl restart nginx`

```
sudo systemctl restart nginx
```

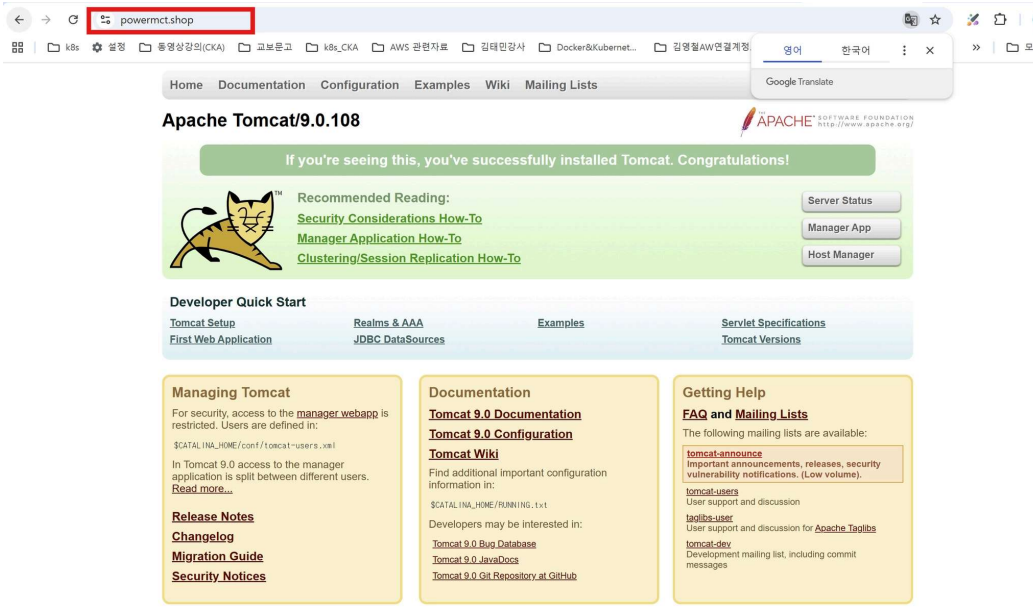
2. 재가동 후 nginx-2C 공인IP 로 접속하면 tomcat 야옹이 화면을 볼 수 있음



3. **이번에는 ALB DNS 정보를 통해서도 tomcat 야옹이 화면을 볼 수 있다.**



4. Route53에 등록했던 각자 도메인으로 연결시 tomcat 야옹이 화면을 볼 수 있다.



07. tomcat 수정 (5개의 jsp 파일을 Bastion-Server에 생성한다.)



1. vi delete.jsp

```
<%@ page language="java" contentType="text/html; charset=UTF-8" p
<%@ page import="java.sql.*" %>
<%@ page import="javax.sql.*" %>
<%@ page import="javax.naming.*" %>
<%!
    // 간단한 HTML 이스케이프 (XSS 방지)
    private String esc(String s) {
        if (s == null) return "";
        return s.replace("&", "&amp;").replace("<", "&lt;")
                .replace(">", "&gt;").replace("\"", "&quot;");
    }
%>
<%
// sypark(2026.06.22) delete.jsp - index.jsp 리스트의 삭제 링크(co
request.setCharacterEncoding("UTF-8");
response.setCharacterEncoding("UTF-8");

String idParam = request.getParameter("id");
int id = -1;
String errorMsg = null;

if (idParam == null) {
```

```
    errorMsg = "잘못된 요청입니다. 게시글 번호가 없습니다.";
} else {
    try {
        id = Integer.parseInt(idParam);
    } catch (Exception ex) {
        errorMsg = "잘못된 게시글 번호입니다.";
    }
}

if (errorMsg == null) {
    Connection conn = null;
    PreparedStatement pstmt = null;
    try {
        Context initContext = new InitialContext();
        Context envContext = (Context) initContext.lookup("java:/
        DataSource ds = (DataSource) envContext.lookup("jdbc/MyDB
        conn = ds.getConnection();

        String sql = "DELETE FROM posts WHERE id = ?";
        pstmt = conn.prepareStatement(sql);
        pstmt.setInt(1, id);
        pstmt.executeUpdate();

        response.sendRedirect("index.jsp");
        return;
    } catch (Exception e) {
        errorMsg = "데이터베이스에 연결할 수 없습니다. (" + e.getMe
    } finally {
        if (pstmt != null) try { pstmt.close(); } catch (Exceptio
        if (conn != null) try { conn.close(); } catch (Exception
    }
}
%>
<!DOCTYPE html>
<html>
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-sc
    <title>삭제 오류 - AWS 3Tier 실습</title>

    <link rel="preconnect" href="https://fonts.googleapis.com">
    <link rel="preconnect" href="https://fonts.gstatic.com" cross
    <link href="https://fonts.googleapis.com/css2?family=Noto+San

    <style>
        :root {
            --accent: #3461ee;
            --accent-hover: #2950d6;
            --text-primary: #18181b;
            --text-secondary: #6b7280;
            --border: #e5e7eb;
```

```
    --bg-page: #ffffff;
    --bg-soft: #f9fafb;
    --error-bg: #fef2f2;
    --error-border: #fecaca;
    --error-text: #b91c1c;
}

* { margin: 0; padding: 0; box-sizing: border-box; }

body {
    font-family: 'Noto Sans KR', -apple-system, BlinkMacS
    background: var(--bg-page);
    color: var(--text-primary);
    padding: 48px 24px 80px;
}

.wrap { max-width: 720px; margin: 0 auto; }

.page-header {
    display: flex;
    justify-content: space-between;
    align-items: flex-start;
    margin-bottom: 28px;
    gap: 16px;
    flex-wrap: wrap;
}

.page-title { font-size: 26px; font-weight: 700; margin-b

.page-subtitle { font-size: 14px; color: var(--text-secon

.btn {
    display: inline-flex;
    align-items: center;
    gap: 6px;
    padding: 10px 18px;
    border-radius: 8px;
    font-size: 14px;
    font-weight: 600;
    text-decoration: none;
    white-space: nowrap;
    border: none;
    cursor: pointer;
    font-family: inherit;
    transition: background .15s ease, border-color .15s e
}

.btn-primary { background: var(--accent); color: #fff; }
.btn-primary:hover { background: var(--accent-hover); }

.btn-secondary {
```

```
            background: #fff;
            color: var(--text-secondary);
            border: 1px solid var(--border);
        }
        .btn-secondary:hover { background: var(--bg-soft); }

        .form-error {
            background: var(--error-bg);
            border: 1px solid var(--error-border);
            color: var(--error-text);
            border-radius: 10px;
            padding: 16px 20px;
            margin-bottom: 20px;
            font-size: 14px;
        }

        .empty-state {
            text-align: center;
            padding: 50px 20px;
            border: 1px solid var(--border);
            border-radius: 10px;
        }

        @media (max-width: 560px) {
            body { padding: 28px 16px 60px; }
            .page-title { font-size: 22px; }
        }
    </style>
</head>
<body>
    <div class="wrap">
        <div class="page-header">
            <div>
                <div class="page-title">삭제 오류</div>
                <div class="page-subtitle">게시글을 삭제하는 중 문
            </div>
            <a href="index.jsp" class="btn btn-secondary">목록으로
        </div>

        <div class="form-error"><%= esc(errorMsg) %></div>

        <div class="empty-state">
            <a href="index.jsp" class="btn btn-primary">목록으로 돌
        </div>
    </div>
</body>
</html>
```

2. **vi edit.jsp**

```jsp
<%@ page language="java" contentType="text/html; charset=UTF-8" p
<%@ page import="java.sql.*" %>
<%@ page import="javax.sql.*" %>
<%@ page import="javax.naming.*" %>
<%!
    // 간단한 HTML 이스케이프 (XSS 방지)
    private String esc(String s) {
        if (s == null) return "";
        return s.replace("&", "&amp;").replace("<", "&lt;")
                .replace(">", "&gt;").replace("\"", "&quot;");
    }
%>
<%
// sypark(2026.06.22) edit.jsp - index.jsp/write.jsp와 동일한 UI 틀
request.setCharacterEncoding("UTF-8");
response.setCharacterEncoding("UTF-8");

String idParam = request.getParameter("id");
int id = -1;
String title = request.getParameter("title");
String author = request.getParameter("author");
String content = request.getParameter("content");

boolean isPost = "POST".equalsIgnoreCase(request.getMethod());
String errorMsg = null;
boolean showForm = true;

if (idParam == null) {
    errorMsg = "잘못된 접근입니다. 게시글 번호가 없습니다.";
    showForm = false;
} else {
    try {
        id = Integer.parseInt(idParam);
    } catch (Exception ex) {
        errorMsg = "잘못된 게시글 번호입니다.";
        showForm = false;
    }
}

if (errorMsg == null && !isPost) {
    // GET: 기존 게시글 내용을 폼에 채워서 보여줌
    Connection conn = null;
    PreparedStatement pstmt = null;
    ResultSet rs = null;
    try {
        Context initContext = new InitialContext();
        Context envContext = (Context) initContext.lookup("java:/
        DataSource ds = (DataSource) envContext.lookup("jdbc/MyDB
        conn = ds.getConnection();
```

```
            String sql = "SELECT title, author, content FROM posts WH
            pstmt = conn.prepareStatement(sql);
            pstmt.setInt(1, id);
            rs = pstmt.executeQuery();
            if (rs.next()) {
                title = rs.getString("title");
                author = rs.getString("author");
                content = rs.getString("content");
            } else {
                errorMsg = "게시글을 찾을 수 없습니다.";
                showForm = false;
            }
        } catch (Exception e) {
            errorMsg = "데이터베이스에 연결할 수 없습니다. (" + e.getMe
            showForm = false;
        } finally {
            if (rs != null) try { rs.close(); } catch (Exception e) {
            if (pstmt != null) try { pstmt.close(); } catch (Exceptio
            if (conn != null) try { conn.close(); } catch (Exception
        }
    }

    if (title == null) title = "";
    if (author == null) author = "";
    if (content == null) content = "";

    if (errorMsg == null && isPost) {
        title = title.trim();
        author = author.trim();
        content = content.trim();

        if (title.isEmpty() || author.isEmpty() || content.isEmpty())
            errorMsg = "제목, 작성자, 내용을 모두 입력해주세요.";
        } else {
            Connection conn = null;
            PreparedStatement pstmt = null;
            try {
                Context initContext = new InitialContext();
                Context envContext = (Context) initContext.lookup("ja
                DataSource ds = (DataSource) envContext.lookup("jdbc/
                conn = ds.getConnection();

                String sql = "UPDATE posts SET title = ?, author = ?,
                pstmt = conn.prepareStatement(sql);
                pstmt.setString(1, title);
                pstmt.setString(2, author);
                pstmt.setString(3, content);
                pstmt.setInt(4, id);
                pstmt.executeUpdate();

                response.sendRedirect("view.jsp?id=" + id);
```

```
            return;
        } catch (Exception e) {
            errorMsg = "데이터베이스에 연결할 수 없습니다. (" + e.g
        } finally {
            if (pstmt != null) try { pstmt.close(); } catch (Exce
            if (conn != null) try { conn.close(); } catch (Except
        }
    }
}

String cancelHref = (id > 0) ? ("view.jsp?id=" + id) : "index.jsp
%>
<!DOCTYPE html>
<html>
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-sc
    <title>글 수정 - AWS 3Tier 실습</title>

    <link rel="preconnect" href="https://fonts.googleapis.com">
    <link rel="preconnect" href="https://fonts.gstatic.com" cross
    <link href="https://fonts.googleapis.com/css2?family=Noto+San

    <style>
        :root {
            --accent: #ef5350;
            --accent-hover: #e23e3b;
            --text-primary: #18181b;
            --text-secondary: #6b7280;
            --text-muted: #9ca3af;
            --border: #e5e7eb;
            --bg-page: #ffffff;
            --bg-soft: #f9fafb;
            --error-bg: #fef2f2;
            --error-border: #fecaca;
            --error-text: #b91c1c;
        }

        * { margin: 0; padding: 0; box-sizing: border-box; }

        body {
            font-family: 'Noto Sans KR', -apple-system, BlinkMacS
            background: var(--bg-page);
            color: var(--text-primary);
            padding: 48px 24px 80px;
        }

        .wrap { max-width: 720px; margin: 0 auto; }

        .page-header {
            display: flex;
```

```
    justify-content: space-between;
    align-items: flex-start;
    margin-bottom: 28px;
    gap: 16px;
    flex-wrap: wrap;
}

.page-title { font-size: 26px; font-weight: 700; margin-b

.page-subtitle { font-size: 14px; color: var(--text-secon

.btn {
    display: inline-flex;
    align-items: center;
    gap: 6px;
    padding: 10px 18px;
    border-radius: 8px;
    font-size: 14px;
    font-weight: 600;
    text-decoration: none;
    white-space: nowrap;
    border: none;
    cursor: pointer;
    font-family: inherit;
    transition: background .15s ease, border-color .15s e
}

.btn-primary { background: var(--accent); color: #fff; }
.btn-primary:hover { background: var(--accent-hover); }

.btn-secondary {
    background: #fff;
    color: var(--text-secondary);
    border: 1px solid var(--border);
}
.btn-secondary:hover { background: var(--bg-soft); }

.form-error {
    background: var(--error-bg);
    border: 1px solid var(--error-border);
    color: var(--error-text);
    border-radius: 10px;
    padding: 16px 20px;
    margin-bottom: 20px;
    font-size: 14px;
}

.form-card {
    border: 1px solid var(--border);
    border-radius: 10px;
    padding: 32px;
```

```
}

.field { margin-bottom: 22px; }
.field:last-of-type { margin-bottom: 0; }

.field label {
    display: block;
    font-size: 13px;
    font-weight: 600;
    color: var(--text-secondary);
    margin-bottom: 8px;
}

.field label .required { color: var(--error-text); margin

.field input[type="text"],
.field textarea {
    width: 100%;
    border: 1px solid var(--border);
    border-radius: 8px;
    padding: 11px 14px;
    font-size: 14px;
    font-family: inherit;
    color: var(--text-primary);
    background: #fff;
}

.field input[type="text"]:focus,
.field textarea:focus {
    outline: none;
    border-color: var(--accent);
}

.field textarea {
    resize: vertical;
    min-height: 280px;
    line-height: 1.6;
}

.row-half {
    display: flex;
    gap: 16px;
}

.row-half .field { flex: 1; }

.form-actions {
    display: flex;
    justify-content: flex-end;
    gap: 8px;
    margin-top: 28px;
```

```
            padding-top: 24px;
            border-top: 1px solid var(--border);
        }

        .empty-state {
            text-align: center;
            padding: 70px 20px;
            color: var(--text-secondary);
            border: 1px solid var(--border);
            border-radius: 10px;
        }

        @media (max-width: 560px) {
            body { padding: 28px 16px 60px; }
            .page-title { font-size: 22px; }
            .form-card { padding: 22px; }
            .row-half { flex-direction: column; gap: 0; }
        }
    </style>
</head>
<body>
    <div class="wrap">
        <div class="page-header">
            <div>
                <div class="page-title">글 수정</div>
                <div class="page-subtitle">게시글 내용을 수정합니다
            </div>
            <a href="index.jsp" class="btn btn-secondary">목록으로
        </div>

        <% if (errorMsg != null) { %>
            <div class="form-error"><%= esc(errorMsg) %></div>
        <% } %>

        <% if (showForm) { %>
        <form class="form-card" action="edit.jsp" method="post">
            <input type="hidden" name="id" value="<%= id %>">
            <div class="row-half">
                <div class="field">
                    <label for="author">작성자<span class="require
                    <input type="text" id="author" name="author"
                </div>
            </div>
            <div class="field">
                <label for="title">제목<span class="required">*</s
                <input type="text" id="title" name="title" value=
            </div>
            <div class="field">
                <label for="content">내용<span class="required">*
                <textarea id="content" name="content" placeholder
            </div>
```

```
            <div class="form-actions">
                <a href="<%= cancelHref %>" class="btn btn-second
                <button type="submit" class="btn btn-primary">수정
            </div>
        </form>
        <% } else { %>
            <div class="empty-state">
                <a href="index.jsp" class="btn btn-primary">목록으
            </div>
        <% } %>
    </div>
</body>
</html>
```

3. **vi index.jsp**

```
<%@ page language="java" contentType="text/html; charset=UTF-8" p
<%@ page import="java.sql.*" %>
<%@ page import="javax.sql.*" %>
<%@ page import="javax.naming.*" %>
<%@ page import="java.text.SimpleDateFormat" %>
<%@ page import="java.net.URLEncoder" %>
<%@ page import="java.net.InetAddress" %>
<%@ page import="java.util.List" %>
<%@ page import="java.util.ArrayList" %>
<%!
    // 간단한 HTML 이스케이프 (XSS 방지)
    private String esc(String s) {
        if (s == null) return "";
        return s.replace("&", "&amp;").replace("<", "&lt;")
                .replace(">", "&gt;").replace("\"", "&quot;");
    }

    // 작성자 아바타 (이니셜 + 고정 색상)
    private static final String[] AVATAR_COLORS = {
        "#f97066", "#fb923c", "#f59e0b", "#34d399", "#22d3ee", "#
    };
    private String avatarColor(String name) {
        if (name == null || name.isEmpty()) return AVATAR_COLORS[
        int idx = Math.abs(name.hashCode()) % AVATAR_COLORS.lengt
        return AVATAR_COLORS[idx];
    }
    private String avatarInitial(String name) {
        if (name == null || name.isEmpty()) return "?";
        return name.substring(0, 1).toUpperCase();
    }
%>
```

```
<%
// sypark(2026.06.22) index.jsp + list.jsp 통합 - 접속 시 바로 목록
// sypark(2026.06.22) DB 조회를 HTML 출력 전으로 이동 - 사이드바 DB
request.setCharacterEncoding("UTF-8");
response.setCharacterEncoding("UTF-8");

String q = request.getParameter("q");
if (q == null) q = "";
q = q.trim();

String sort = request.getParameter("sort");
if (sort == null || !sort.equals("old")) sort = "new";

int curPage = 1;
String pageParam = request.getParameter("page");
if (pageParam != null) {
    try { curPage = Integer.parseInt(pageParam); } catch (Exceptio
}
if (curPage < 1) curPage = 1;

final int pageSize = 10;
int totalCount = 0;
int totalPages = 1;

boolean dbError = false;
String dbErrorMsg = "";
List<Object[]> postRows = new ArrayList<Object[]>(); // {id, titl

String qEncoded = "";
try { qEncoded = URLEncoder.encode(q, "UTF-8"); } catch (Exceptio

// 현재 접속(서비스) 서버 IP
String serverIp;
try {
    serverIp = InetAddress.getLocalHost().getHostAddress();
} catch (Exception ex) {
    serverIp = request.getServerName();
}

Connection conn = null;
PreparedStatement countStmt = null;
PreparedStatement listStmt = null;
ResultSet countRs = null;
ResultSet rs = null;

try {
    Context initContext = new InitialContext();
    Context envContext = (Context) initContext.lookup("java:/comp
    DataSource ds = (DataSource) envContext.lookup("jdbc/MyDB");
    conn = ds.getConnection();
```

```
        boolean hasSearch = !q.isEmpty();
        String whereClause = hasSearch ? " WHERE (title LIKE ? OR con
        String orderClause = "old".equals(sort) ? " ORDER BY created_

        // 전체 건수 조회 (페이지네이션 계산용)
        String countSql = "SELECT COUNT(*) AS cnt FROM posts" + where
        countStmt = conn.prepareStatement(countSql);
        if (hasSearch) {
            countStmt.setString(1, "%" + q + "%");
            countStmt.setString(2, "%" + q + "%");
            countStmt.setString(3, "%" + q + "%");
        }
        countRs = countStmt.executeQuery();
        if (countRs.next()) totalCount = countRs.getInt("cnt");

        totalPages = (int) Math.ceil(totalCount / (double) pageSize);
        if (totalPages < 1) totalPages = 1;
        if (curPage > totalPages) curPage = totalPages;

        int offset = (curPage - 1) * pageSize;

        String listSql = "SELECT id, title, author, created_at FROM p
                       + orderClause + " LIMIT ? OFFSET ?";
        listStmt = conn.prepareStatement(listSql);
        int idx = 1;
        if (hasSearch) {
            listStmt.setString(idx++, "%" + q + "%");
            listStmt.setString(idx++, "%" + q + "%");
            listStmt.setString(idx++, "%" + q + "%");
        }
        listStmt.setInt(idx++, pageSize);
        listStmt.setInt(idx++, offset);
        rs = listStmt.executeQuery();

        while (rs.next()) {
            postRows.add(new Object[] {
                rs.getInt("id"), rs.getString("title"), rs.getString(
            });
        }
    } catch (Exception e) {
        dbError = true;
        dbErrorMsg = e.getMessage();
    } finally {
        if (rs != null) try { rs.close(); } catch (Exception e) {}
        if (countRs != null) try { countRs.close(); } catch (Exceptio
        if (listStmt != null) try { listStmt.close(); } catch (Except
        if (countStmt != null) try { countStmt.close(); } catch (Exce
        if (conn != null) try { conn.close(); } catch (Exception e) {
    }
    %>
    <!DOCTYPE html>
```

```
<html>
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-sc
    <title>AWS 3Tier 실습</title>

    <link rel="preconnect" href="https://fonts.googleapis.com">
    <link rel="preconnect" href="https://fonts.gstatic.com" cross
    <link href="https://fonts.googleapis.com/css2?family=Noto+San

    <style>
        :root {
            --accent: #ef5350;
            --accent-hover: #e23e3b;
            --text-primary: #18181b;
            --text-secondary: #6b7280;
            --text-muted: #9ca3af;
            --border: #e5e7eb;
            --border-light: #f1f3f5;
            --bg-page: #ffffff;
            --bg-soft: #f9fafb;
            --error-bg: #fef2f2;
            --error-border: #fecaca;
            --error-text: #b91c1c;
            --sidebar-from: #f3614f;
            --sidebar-to: #e2493a;
            --sidebar-text: rgba(255, 255, 255, .94);
            --sidebar-text-dim: rgba(255, 255, 255, .68);
            --sidebar-card-bg: rgba(255, 255, 255, .13);
            --sidebar-card-border: rgba(255, 255, 255, .2);
        }

        * { margin: 0; padding: 0; box-sizing: border-box; }

        html, body { height: 100%; }

        body {
            font-family: 'Noto Sans KR', -apple-system, BlinkMacS
            background: var(--bg-page);
            color: var(--text-primary);
            display: flex;
            align-items: stretch;
            min-height: 100vh;
        }

        /* 좌측 사이드바 - 시스템 구조 설명 */
        .sidebar {
            width: 300px;
            flex-shrink: 0;
            background: linear-gradient(180deg, var(--sidebar-fro
            color: var(--sidebar-text);
```

```
    padding: 36px 28px;
    display: flex;
    flex-direction: column;
    position: sticky;
    top: 0;
    height: 100vh;
    overflow-y: auto;
}

.sidebar-brand {
    font-size: 21px;
    font-weight: 700;
    margin-bottom: 22px;
    letter-spacing: -.2px;
}

.sidebar-brand .dim { font-weight: 400; opacity: .8; }

.sidebar-intro {
    font-size: 13px;
    line-height: 1.65;
    color: var(--sidebar-text-dim);
    margin-bottom: 26px;
    padding-bottom: 26px;
    border-bottom: 1px solid rgba(255, 255, 255, .18);
}

.sidebar-heading {
    font-size: 11px;
    font-weight: 700;
    letter-spacing: .06em;
    text-transform: uppercase;
    color: var(--sidebar-text-dim);
    margin-bottom: 14px;
}

.tier-flow { display: flex; flex-direction: column; }

.tier-card {
    background: var(--sidebar-card-bg);
    border: 1px solid var(--sidebar-card-border);
    border-radius: 10px;
    padding: 13px 14px;
}

.tier-name {
    font-size: 13px;
    font-weight: 700;
    margin-bottom: 3px;
}
```

```
.tier-tech {
    font-size: 12px;
    color: var(--sidebar-text-dim);
    margin-bottom: 8px;
}

.tier-status {
    display: inline-flex;
    align-items: center;
    gap: 5px;
    font-size: 11px;
    font-weight: 600;
    color: var(--sidebar-text);
}

.tier-status .dot {
    width: 6px;
    height: 6px;
    border-radius: 50%;
    background: #4ade80;
    display: inline-block;
}

.tier-card-error {
    background: rgba(0, 0, 0, .15);
    border-color: rgba(251, 191, 36, .55);
}

.tier-status.status-error { color: #fde68a; }

.tier-status .dot-error {
    background: #fbbf24;
    box-shadow: 0 0 0 3px rgba(251, 191, 36, .25);
}

.tier-arrow {
    text-align: center;
    font-size: 14px;
    color: var(--sidebar-text-dim);
    padding: 4px 0;
}

.sidebar-flow-desc {
    font-size: 12.5px;
    line-height: 1.7;
    color: var(--sidebar-text-dim);
    margin-top: 18px;
}

.sidebar-footer {
    margin-top: auto;
```

```
    padding-top: 20px;
    border-top: 1px solid rgba(255, 255, 255, .18);
    font-size: 11.5px;
    line-height: 1.7;
    color: var(--sidebar-text-dim);
}

/* 메인 영역 */
.main { flex: 1; min-width: 0; padding: 48px 40px 80px; }

.wrap { max-width: 1000px; margin: 0 auto; }

/* 헤더 */
.page-header {
    display: flex;
    justify-content: space-between;
    align-items: flex-start;
    margin-bottom: 28px;
    gap: 16px;
    flex-wrap: wrap;
}

.page-title { font-size: 26px; font-weight: 700; margin-b

.page-subtitle { font-size: 14px; color: var(--text-secon

.btn-write {
    display: inline-flex;
    align-items: center;
    gap: 6px;
    background: var(--accent);
    color: #fff;
    padding: 10px 18px;
    border-radius: 8px;
    font-size: 14px;
    font-weight: 600;
    text-decoration: none;
    white-space: nowrap;
    transition: background .15s ease;
}

.btn-write:hover { background: var(--accent-hover); }

.btn-write svg { width: 15px; height: 15px; }

/* 검색바 */
.toolbar {
    display: flex;
    gap: 10px;
    margin-bottom: 18px;
    flex-wrap: wrap;
```

```
}

.select-cat,
.select-sort,
.search-input {
    border: 1px solid var(--border);
    border-radius: 8px;
    padding: 10px 14px;
    font-size: 14px;
    font-family: inherit;
    color: var(--text-primary);
    background: #fff;
}

.select-cat, .select-sort { min-width: 110px; cursor: poi
.select-sort:focus { outline: none; border-color: var(--a

.search-form {
    display: flex;
    gap: 8px;
    flex: 1;
    min-width: 240px;
}

.search-input { flex: 1; }

.search-input:focus,
.select-cat:focus { outline: none; border-color: var(--ac

.btn-search {
    display: inline-flex;
    align-items: center;
    justify-content: center;
    width: 42px;
    border: 1px solid var(--border);
    border-radius: 8px;
    background: #fff;
    cursor: pointer;
    transition: background .15s ease;
}

.btn-search:hover { background: var(--bg-soft); }

.btn-search svg { width: 17px; height: 17px; stroke: var(

/* 에러 배너 */
.db-error {
    background: var(--error-bg);
    border: 1px solid var(--error-border);
    color: var(--error-text);
    border-radius: 10px;
```

```
    padding: 22px 24px;
    margin-top: 8px;
}

.db-error .title { font-weight: 700; font-size: 16px; mar

.db-error .desc { font-size: 13px; color: #991b1b; line-h

/* 테이블 */
.table-wrap {
    border: 1px solid var(--border);
    border-radius: 10px;
    overflow-x: auto;
}

table { width: 100%; border-collapse: collapse; min-width

thead th {
    background: var(--bg-soft);
    color: var(--text-secondary);
    font-size: 13px;
    font-weight: 600;
    text-align: center;
    padding: 14px 10px;
    border-bottom: 1px solid var(--border);
    white-space: nowrap;
}

th.col-title, td.col-title, th.col-author { text-align: l

tbody td {
    padding: 14px 10px;
    font-size: 14px;
    text-align: center;
    border-bottom: 1px solid var(--border-light);
    color: var(--text-primary);
}

tbody tr:last-child td { border-bottom: none; }

tbody tr:hover { background: var(--bg-soft); }

td.col-no { color: var(--text-muted); width: 60px; }

td.col-title {
    max-width: 0;
    width: 100%;
    overflow: hidden;
    text-overflow: ellipsis;
    white-space: nowrap;
}
```

```
td.col-title a {
    color: var(--text-primary);
    text-decoration: none;
    font-weight: 500;
}

td.col-title a:hover { color: var(--accent); text-decorat

td.col-author {
    width: 150px;
    color: var(--text-secondary);
    text-align: left;
}

.author-cell {
    display: inline-flex;
    align-items: center;
    gap: 8px;
    max-width: 100%;
}

.author-name {
    white-space: nowrap;
    overflow: hidden;
    text-overflow: ellipsis;
}

.avatar {
    display: inline-flex;
    align-items: center;
    justify-content: center;
    width: 24px;
    height: 24px;
    flex-shrink: 0;
    border-radius: 50%;
    color: #fff;
    font-size: 11px;
    font-weight: 700;
}

td.col-date { width: 110px; color: var(--text-secondary);

td.col-manage { width: 120px; font-size: 13px; white-spac

td.col-manage a {
    color: var(--text-muted);
    text-decoration: none;
}

td.col-manage a:hover { color: var(--accent); text-decora
```

```
td.col-manage a.del:hover { color: #d92d20; }

td.col-manage .sep { color: var(--border); margin: 0 6px;

/* 빈 상태 */
.empty-state {
    text-align: center;
    padding: 70px 20px;
    color: var(--text-secondary);
}

.empty-state .title { font-size: 16px; font-weight: 600;

.empty-state .desc { font-size: 13px; }

/* 페이지네이션 */
.pagination {
    display: flex;
    justify-content: center;
    align-items: center;
    gap: 4px;
    margin-top: 28px;
}

.pagination a, .pagination span {
    display: inline-flex;
    align-items: center;
    justify-content: center;
    min-width: 32px;
    height: 32px;
    padding: 0 6px;
    border-radius: 7px;
    font-size: 13px;
    text-decoration: none;
    color: var(--text-secondary);
}

.pagination a:hover { background: var(--bg-soft); }

.pagination .current {
    background: var(--accent);
    color: #fff;
    font-weight: 600;
}

.pagination .ellipsis { color: var(--text-muted); }

.pagination .nav-arrow { color: var(--text-muted); }

.pagination .nav-arrow.disabled { opacity: .35; pointer-e
```

```
        @media (max-width: 860px) {
            body { flex-direction: column; }
            .sidebar {
                width: 100%;
                height: auto;
                position: static;
                padding: 28px 24px;
            }
            .tier-flow { flex-direction: row; gap: 10px; overflow
            .tier-arrow { display: none; }
            .sidebar-flow-desc { display: none; }
        }

        @media (max-width: 560px) {
            .main { padding: 28px 16px 60px; }
            .page-title { font-size: 22px; }
        }
    </style>
</head>
<body>
    <aside class="sidebar">
        <div class="sidebar-brand">3Tier<span class="dim"> Board<

        <div class="sidebar-intro">
            AWS 3-Tier 아키텍처 실습용으로 만든 게시판입니다.
            아래는 이 서비스가 실제로 어떤 구조로 요청을 처리하는지
        </div>

        <div class="sidebar-heading">시스템 아키텍처</div>
        <div class="tier-flow">
            <div class="tier-card">
                <div class="tier-name">Web Tier</div>
                <div class="tier-tech">Nginx</div>
                <div class="tier-status"><span class="dot"></span
            </div>
            <div class="tier-arrow">↓</div>
            <div class="tier-card">
                <div class="tier-name">Application Tier</div>
                <div class="tier-tech">Apache Tomcat 9.0</div>
                <div class="tier-status"><span class="dot"></span
            </div>
            <div class="tier-arrow">↓</div>
            <div class="tier-card<%= dbError ? " tier-card-error"
                <div class="tier-name">Database Tier</div>
                <div class="tier-tech">AWS RDS MySQL</div>
                <div class="tier-status<%= dbError ? " status-err
                    <span class="dot<%= dbError ? " dot-error" :
                </div>
            </div>
        </div>
```

```
    <div class="sidebar-flow-desc">
        클라이언트 요청은 Nginx가 받아 Tomcat으로 전달하고, Tom
        AWS RDS MySQL과 통신해 데이터를 저장하고 조회합니다.
    </div>

    <div class="sidebar-footer">
        접속서버: <%= serverIp %><br>
        <%= request.getServerName() %>:<%= request.getServerP
        <%= new java.text.SimpleDateFormat("yyyy.MM.dd HH:mm"
    </div>
</aside>

<main class="main">
<div class="wrap">
    <div class="page-header">
        <div>
            <div class="page-title">AWS 3Tier 실습</div>
            <div class="page-subtitle">AWS 3Tier 아키텍처 실습
        </div>
        <a href="write.jsp" class="btn-write">
            <svg viewBox="0 0 24 24" fill="none" stroke="curr
            글쓰기
        </a>
    </div>

    <form class="toolbar" action="index.jsp" method="get">
        <select class="select-cat" disabled>
            <option>전체 게시글</option>
        </select>
        <select class="select-sort" name="sort" onchange="thi
            <option value="new" <%= "new".equals(sort) ? "sel
            <option value="old" <%= "old".equals(sort) ? "sel
        </select>
        <div class="search-form">
            <input
                type="text"
                name="q"
                class="search-input"
                placeholder="제목, 내용, 작성자 검색..."
                value="<%= esc(q) %>"
            />
            <button type="submit" class="btn-search" aria-lab
                <svg viewBox="0 0 24 24" fill="none" stroke-w
            </button>
        </div>
    </form>

    <% if (dbError) { %>
        <div class="db-error">
            <div class="title">데이터베이스에 연결할 수 없습니다
```

```
            <div class="desc"><%= esc(dbErrorMsg) %></div>
        </div>
    <% } else { %>

    <div class="table-wrap">
        <table>
            <thead>
                <tr>
                    <th style="width:60px;">번호</th>
                    <th class="col-title">제목</th>
                    <th style="width:150px;" class="col-autho
                    <th style="width:110px;">작성일</th>
                    <th style="width:120px;">관리</th>
                </tr>
            </thead>
            <tbody>
            <%
            SimpleDateFormat sdf = new SimpleDateFormat("yyyy
            for (Object[] row : postRows) {
                int id = (Integer) row[0];
                String rawAuthor = (String) row[2];
                String title = esc((String) row[1]);
                String author = esc(rawAuthor);
                java.sql.Timestamp ts = (java.sql.Timestamp)
                String dateStr = ts != null ? sdf.format(ts)
            %>
                <tr>
                    <td class="col-no"><%= id %></td>
                    <td class="col-title" title="<%= title %>
                        <a href="view.jsp?id=<%= id %>"><%= t
                    </td>
                    <td class="col-author">
                        <span class="author-cell">
                            <span class="avatar" style="backg
                            <span class="author-name"><%= aut
                        </span>
                    </td>
                    <td class="col-date"><%= dateStr %></td>
                    <td class="col-manage">
                        <a href="edit.jsp?id=<%= id %>">수정</
                    </td>
                </tr>
            <%
            }
            if (postRows.isEmpty()) {
            %>
                <tr>
                    <td colspan="5">
                        <div class="empty-state">
                            <div class="title"><%= q.isEmpty(
                            <div class="desc">
```

```
                                <% if (q.isEmpty()) { %>
                                    첫 번째 게시글을 작성해보세
                                <% } else { %>
                                    '<%= esc(q) %>'에 대한 검색
                                <% } %>
                            </div>
                        </div>
                    </td>
                </tr>
            <%
            }
            %>
            </tbody>
        </table>
    </div>

    <%
    if (totalPages > 1) {
        int windowStart, windowEnd;
        if (totalPages <= 7) {
            windowStart = 1; windowEnd = totalPages;
        } else if (curPage <= 4) {
            windowStart = 1; windowEnd = 5;
        } else if (curPage >= totalPages - 3) {
            windowStart = totalPages - 4; windowEnd = totalPa
        } else {
            windowStart = curPage - 2; windowEnd = curPage +
        }
        String qSuffix = q.isEmpty() ? "" : ("&q=" + qEncoded
        qSuffix += "old".equals(sort) ? "&sort=old" : "";
    %>
    <div class="pagination">
        <a class="nav-arrow<%= curPage <= 1 ? " disabled" : "
        <a class="nav-arrow<%= curPage <= 1 ? " disabled" : "
        <%
        if (windowStart > 1) {
        %>
            <a href="index.jsp?page=1<%= qSuffix %>">1</a>
            <% if (windowStart > 2) { %><span class="ellipsis
        <%
        }
        for (int p = windowStart; p <= windowEnd; p++) {
            if (p == curPage) {
        %>
                <span class="current"><%= p %></span>
        <%
            } else {
        %>
                <a href="index.jsp?page=<%= p %><%= qSuffix %
        <%
            }
```

```
            }
            if (windowEnd < totalPages) {
                if (windowEnd < totalPages - 1) {
            %>
                    <span class="ellipsis">…</span>
            <%
                }
            %>
                <a href="index.jsp?page=<%= totalPages %><%= qSuf
            <%
            }
            %>
            <a class="nav-arrow<%= curPage >= totalPages ? " disa
        </div>
        <%
        }
        %>

        <% } %>
    </div>
    </main>
</body>
</html>
```

4. vi view.jsp

```
<%@ page language="java" contentType="text/html; charset=UTF-8" p
<%@ page import="java.sql.*" %>
<%@ page import="javax.sql.*" %>
<%@ page import="javax.naming.*" %>
<%@ page import="java.text.SimpleDateFormat" %>
<%@ page import="java.net.URLEncoder" %>
<%@ page import="java.net.InetAddress" %>
<%@ page import="java.util.List" %>
<%@ page import="java.util.ArrayList" %>
<%!
    // 간단한 HTML 이스케이프 (XSS 방지)
    private String esc(String s) {
        if (s == null) return "";
        return s.replace("&", "&amp;").replace("<", "&lt;")
                .replace(">", "&gt;").replace("\"", "&quot;");
    }

    // 작성자 아바타 (이니셜 + 고정 색상)
    private static final String[] AVATAR_COLORS = {
        "#f97066", "#fb923c", "#f59e0b", "#34d399", "#22d3ee", "#
    };
    private String avatarColor(String name) {
```

```
        if (name == null || name.isEmpty()) return AVATAR_COLORS[
        int idx = Math.abs(name.hashCode()) % AVATAR_COLORS.lengt
        return AVATAR_COLORS[idx];
    }
    private String avatarInitial(String name) {
        if (name == null || name.isEmpty()) return "?";
        return name.substring(0, 1).toUpperCase();
    }
%>
<%
// sypark(2026.06.22) index.jsp + list.jsp 통합 - 접속 시 바로 목록
// sypark(2026.06.22) DB 조회를 HTML 출력 전으로 이동 - 사이드바 DB
request.setCharacterEncoding("UTF-8");
response.setCharacterEncoding("UTF-8");

String q = request.getParameter("q");
if (q == null) q = "";
q = q.trim();

String sort = request.getParameter("sort");
if (sort == null || !sort.equals("old")) sort = "new";

int curPage = 1;
String pageParam = request.getParameter("page");
if (pageParam != null) {
    try { curPage = Integer.parseInt(pageParam); } catch (Exceptio
}
if (curPage < 1) curPage = 1;

final int pageSize = 10;
int totalCount = 0;
int totalPages = 1;

boolean dbError = false;
String dbErrorMsg = "";
List<Object[]> postRows = new ArrayList<Object[]>(); // {id, titl

String qEncoded = "";
try { qEncoded = URLEncoder.encode(q, "UTF-8"); } catch (Exceptio

// 현재 접속(서비스) 서버 IP
String serverIp;
try {
    serverIp = InetAddress.getLocalHost().getHostAddress();
} catch (Exception ex) {
    serverIp = request.getServerName();
}

Connection conn = null;
PreparedStatement countStmt = null;
PreparedStatement listStmt = null;
```

```
ResultSet countRs = null;
ResultSet rs = null;

try {
    Context initContext = new InitialContext();
    Context envContext = (Context) initContext.lookup("java:/comp
    DataSource ds = (DataSource) envContext.lookup("jdbc/MyDB");
    conn = ds.getConnection();

    boolean hasSearch = !q.isEmpty();
    String whereClause = hasSearch ? " WHERE (title LIKE ? OR con
    String orderClause = "old".equals(sort) ? " ORDER BY created_

    // 전체 건수 조회 (페이지네이션 계산용)
    String countSql = "SELECT COUNT(*) AS cnt FROM posts" + where
    countStmt = conn.prepareStatement(countSql);
    if (hasSearch) {
        countStmt.setString(1, "%" + q + "%");
        countStmt.setString(2, "%" + q + "%");
        countStmt.setString(3, "%" + q + "%");
    }
    countRs = countStmt.executeQuery();
    if (countRs.next()) totalCount = countRs.getInt("cnt");

    totalPages = (int) Math.ceil(totalCount / (double) pageSize);
    if (totalPages < 1) totalPages = 1;
    if (curPage > totalPages) curPage = totalPages;

    int offset = (curPage - 1) * pageSize;

    String listSql = "SELECT id, title, author, created_at FROM p
                   + orderClause + " LIMIT ? OFFSET ?";
    listStmt = conn.prepareStatement(listSql);
    int idx = 1;
    if (hasSearch) {
        listStmt.setString(idx++, "%" + q + "%");
        listStmt.setString(idx++, "%" + q + "%");
        listStmt.setString(idx++, "%" + q + "%");
    }
    listStmt.setInt(idx++, pageSize);
    listStmt.setInt(idx++, offset);
    rs = listStmt.executeQuery();

    while (rs.next()) {
        postRows.add(new Object[] {
            rs.getInt("id"), rs.getString("title"), rs.getString(
        });
    }
} catch (Exception e) {
    dbError = true;
    dbErrorMsg = e.getMessage();
```

```
} finally {
    if (rs != null) try { rs.close(); } catch (Exception e) {}
    if (countRs != null) try { countRs.close(); } catch (Exceptio
    if (listStmt != null) try { listStmt.close(); } catch (Except
    if (countStmt != null) try { countStmt.close(); } catch (Exce
    if (conn != null) try { conn.close(); } catch (Exception e) {
}
%>
<!DOCTYPE html>
<html>
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-sc
    <title>AWS 3Tier 실습</title>

    <link rel="preconnect" href="https://fonts.googleapis.com">
    <link rel="preconnect" href="https://fonts.gstatic.com" cross
    <link href="https://fonts.googleapis.com/css2?family=Noto+San

    <style>
        :root {
            --accent: #ef5350;
            --accent-hover: #e23e3b;
            --text-primary: #18181b;
            --text-secondary: #6b7280;
            --text-muted: #9ca3af;
            --border: #e5e7eb;
            --border-light: #f1f3f5;
            --bg-page: #ffffff;
            --bg-soft: #f9fafb;
            --error-bg: #fef2f2;
            --error-border: #fecaca;
            --error-text: #b91c1c;
            --sidebar-from: #f3614f;
            --sidebar-to: #e2493a;
            --sidebar-text: rgba(255, 255, 255, .94);
            --sidebar-text-dim: rgba(255, 255, 255, .68);
            --sidebar-card-bg: rgba(255, 255, 255, .13);
            --sidebar-card-border: rgba(255, 255, 255, .2);
        }

        * { margin: 0; padding: 0; box-sizing: border-box; }

        html, body { height: 100%; }

        body {
            font-family: 'Noto Sans KR', -apple-system, BlinkMacS
            background: var(--bg-page);
            color: var(--text-primary);
            display: flex;
            align-items: stretch;
```

```
    min-height: 100vh;
}

/* 좌측 사이드바 - 시스템 구조 설명 */
.sidebar {
    width: 300px;
    flex-shrink: 0;
    background: linear-gradient(180deg, var(--sidebar-fro
    color: var(--sidebar-text);
    padding: 36px 28px;
    display: flex;
    flex-direction: column;
    position: sticky;
    top: 0;
    height: 100vh;
    overflow-y: auto;
}

.sidebar-brand {
    font-size: 21px;
    font-weight: 700;
    margin-bottom: 22px;
    letter-spacing: -.2px;
}

.sidebar-brand .dim { font-weight: 400; opacity: .8; }

.sidebar-intro {
    font-size: 13px;
    line-height: 1.65;
    color: var(--sidebar-text-dim);
    margin-bottom: 26px;
    padding-bottom: 26px;
    border-bottom: 1px solid rgba(255, 255, 255, .18);
}

.sidebar-heading {
    font-size: 11px;
    font-weight: 700;
    letter-spacing: .06em;
    text-transform: uppercase;
    color: var(--sidebar-text-dim);
    margin-bottom: 14px;
}

.tier-flow { display: flex; flex-direction: column; }

.tier-card {
    background: var(--sidebar-card-bg);
    border: 1px solid var(--sidebar-card-border);
    border-radius: 10px;
```

```
    padding: 13px 14px;
}

.tier-name {
    font-size: 13px;
    font-weight: 700;
    margin-bottom: 3px;
}

.tier-tech {
    font-size: 12px;
    color: var(--sidebar-text-dim);
    margin-bottom: 8px;
}

.tier-status {
    display: inline-flex;
    align-items: center;
    gap: 5px;
    font-size: 11px;
    font-weight: 600;
    color: var(--sidebar-text);
}

.tier-status .dot {
    width: 6px;
    height: 6px;
    border-radius: 50%;
    background: #4ade80;
    display: inline-block;
}

.tier-card-error {
    background: rgba(0, 0, 0, .15);
    border-color: rgba(251, 191, 36, .55);
}

.tier-status.status-error { color: #fde68a; }

.tier-status .dot-error {
    background: #fbbf24;
    box-shadow: 0 0 0 3px rgba(251, 191, 36, .25);
}

.tier-arrow {
    text-align: center;
    font-size: 14px;
    color: var(--sidebar-text-dim);
    padding: 4px 0;
}
```

```css
.sidebar-flow-desc {
    font-size: 12.5px;
    line-height: 1.7;
    color: var(--sidebar-text-dim);
    margin-top: 18px;
}

.sidebar-footer {
    margin-top: auto;
    padding-top: 20px;
    border-top: 1px solid rgba(255, 255, 255, .18);
    font-size: 11.5px;
    line-height: 1.7;
    color: var(--sidebar-text-dim);
}

/* 메인 영역 */
.main { flex: 1; min-width: 0; padding: 48px 40px 80px; }

.wrap { max-width: 1000px; margin: 0 auto; }

/* 헤더 */
.page-header {
    display: flex;
    justify-content: space-between;
    align-items: flex-start;
    margin-bottom: 28px;
    gap: 16px;
    flex-wrap: wrap;
}

.page-title { font-size: 26px; font-weight: 700; margin-b

.page-subtitle { font-size: 14px; color: var(--text-secon

.btn-write {
    display: inline-flex;
    align-items: center;
    gap: 6px;
    background: var(--accent);
    color: #fff;
    padding: 10px 18px;
    border-radius: 8px;
    font-size: 14px;
    font-weight: 600;
    text-decoration: none;
    white-space: nowrap;
    transition: background .15s ease;
}

.btn-write:hover { background: var(--accent-hover); }
```

```
.btn-write svg { width: 15px; height: 15px; }

/* 검색바 */
.toolbar {
    display: flex;
    gap: 10px;
    margin-bottom: 18px;
    flex-wrap: wrap;
}

.select-cat,
.select-sort,
.search-input {
    border: 1px solid var(--border);
    border-radius: 8px;
    padding: 10px 14px;
    font-size: 14px;
    font-family: inherit;
    color: var(--text-primary);
    background: #fff;
}

.select-cat, .select-sort { min-width: 110px; cursor: poi
.select-sort:focus { outline: none; border-color: var(--a

.search-form {
    display: flex;
    gap: 8px;
    flex: 1;
    min-width: 240px;
}

.search-input { flex: 1; }

.search-input:focus,
.select-cat:focus { outline: none; border-color: var(--ac

.btn-search {
    display: inline-flex;
    align-items: center;
    justify-content: center;
    width: 42px;
    border: 1px solid var(--border);
    border-radius: 8px;
    background: #fff;
    cursor: pointer;
    transition: background .15s ease;
}

.btn-search:hover { background: var(--bg-soft); }
```

```
.btn-search svg { width: 17px; height: 17px; stroke: var(

/* 에러 배너 */
.db-error {
    background: var(--error-bg);
    border: 1px solid var(--error-border);
    color: var(--error-text);
    border-radius: 10px;
    padding: 22px 24px;
    margin-top: 8px;
}

.db-error .title { font-weight: 700; font-size: 16px; mar

.db-error .desc { font-size: 13px; color: #991b1b; line-h

/* 테이블 */
.table-wrap {
    border: 1px solid var(--border);
    border-radius: 10px;
    overflow-x: auto;
}

table { width: 100%; border-collapse: collapse; min-width

thead th {
    background: var(--bg-soft);
    color: var(--text-secondary);
    font-size: 13px;
    font-weight: 600;
    text-align: center;
    padding: 14px 10px;
    border-bottom: 1px solid var(--border);
    white-space: nowrap;
}

th.col-title, td.col-title, th.col-author { text-align: l

tbody td {
    padding: 14px 10px;
    font-size: 14px;
    text-align: center;
    border-bottom: 1px solid var(--border-light);
    color: var(--text-primary);
}

tbody tr:last-child td { border-bottom: none; }

tbody tr:hover { background: var(--bg-soft); }
```

```
td.col-no { color: var(--text-muted); width: 60px; }

td.col-title {
    max-width: 0;
    width: 100%;
    overflow: hidden;
    text-overflow: ellipsis;
    white-space: nowrap;
}

td.col-title a {
    color: var(--text-primary);
    text-decoration: none;
    font-weight: 500;
}

td.col-title a:hover { color: var(--accent); text-decorat

td.col-author {
    width: 150px;
    color: var(--text-secondary);
    text-align: left;
}

.author-cell {
    display: inline-flex;
    align-items: center;
    gap: 8px;
    max-width: 100%;
}

.author-name {
    white-space: nowrap;
    overflow: hidden;
    text-overflow: ellipsis;
}

.avatar {
    display: inline-flex;
    align-items: center;
    justify-content: center;
    width: 24px;
    height: 24px;
    flex-shrink: 0;
    border-radius: 50%;
    color: #fff;
    font-size: 11px;
    font-weight: 700;
}

td.col-date { width: 110px; color: var(--text-secondary);
```

```
td.col-manage { width: 120px; font-size: 13px; white-spac

td.col-manage a {
    color: var(--text-muted);
    text-decoration: none;
}

td.col-manage a:hover { color: var(--accent); text-decora

td.col-manage a.del:hover { color: #d92d20; }

td.col-manage .sep { color: var(--border); margin: 0 6px;

/* 빈 상태 */
.empty-state {
    text-align: center;
    padding: 70px 20px;
    color: var(--text-secondary);
}

.empty-state .title { font-size: 16px; font-weight: 600;

.empty-state .desc { font-size: 13px; }

/* 페이지네이션 */
.pagination {
    display: flex;
    justify-content: center;
    align-items: center;
    gap: 4px;
    margin-top: 28px;
}

.pagination a, .pagination span {
    display: inline-flex;
    align-items: center;
    justify-content: center;
    min-width: 32px;
    height: 32px;
    padding: 0 6px;
    border-radius: 7px;
    font-size: 13px;
    text-decoration: none;
    color: var(--text-secondary);
}

.pagination a:hover { background: var(--bg-soft); }

.pagination .current {
    background: var(--accent);
```

```
            color: #fff;
            font-weight: 600;
        }

        .pagination .ellipsis { color: var(--text-muted); }

        .pagination .nav-arrow { color: var(--text-muted); }

        .pagination .nav-arrow.disabled { opacity: .35; pointer-e

        @media (max-width: 860px) {
            body { flex-direction: column; }
            .sidebar {
                width: 100%;
                height: auto;
                position: static;
                padding: 28px 24px;
            }
            .tier-flow { flex-direction: row; gap: 10px; overflow
            .tier-arrow { display: none; }
            .sidebar-flow-desc { display: none; }
        }

        @media (max-width: 560px) {
            .main { padding: 28px 16px 60px; }
            .page-title { font-size: 22px; }
        }
    </style>
</head>
<body>
    <aside class="sidebar">
        <div class="sidebar-brand">3Tier<span class="dim"> Board<

        <div class="sidebar-intro">
            AWS 3-Tier 아키텍처 실습용으로 만든 게시판입니다.
            아래는 이 서비스가 실제로 어떤 구조로 요청을 처리하는지
        </div>

        <div class="sidebar-heading">시스템 아키텍처</div>
        <div class="tier-flow">
            <div class="tier-card">
                <div class="tier-name">Web Tier</div>
                <div class="tier-tech">Nginx</div>
                <div class="tier-status"><span class="dot"></span
            </div>
            <div class="tier-arrow">↓</div>
            <div class="tier-card">
                <div class="tier-name">Application Tier</div>
                <div class="tier-tech">Apache Tomcat 9.0</div>
                <div class="tier-status"><span class="dot"></span
            </div>
```

```
            <div class="tier-arrow">↓</div>
            <div class="tier-card<%= dbError ? " tier-card-error"
                <div class="tier-name">Database Tier</div>
                <div class="tier-tech">AWS RDS MySQL</div>
                <div class="tier-status<%= dbError ? " status-err
                    <span class="dot<%= dbError ? " dot-error" :
                </div>
            </div>
        </div>

        <div class="sidebar-flow-desc">
            클라이언트 요청은 Nginx가 받아 Tomcat으로 전달하고, Tom
            AWS RDS MySQL과 통신해 데이터를 저장하고 조회합니다.
        </div>

        <div class="sidebar-footer">
            접속서버: <%= serverIp %><br>
            <%= request.getServerName() %>:<%= request.getServerP
            <%= new java.text.SimpleDateFormat("yyyy.MM.dd HH:mm"
        </div>
    </aside>

    <main class="main">
    <div class="wrap">
        <div class="page-header">
            <div>
                <div class="page-title">AWS 3Tier 실습</div>
                <div class="page-subtitle">AWS 3Tier 아키텍처 실습
            </div>
            <a href="write.jsp" class="btn-write">
                <svg viewBox="0 0 24 24" fill="none" stroke="curr
                글쓰기
            </a>
        </div>

        <form class="toolbar" action="index.jsp" method="get">
            <select class="select-cat" disabled>
                <option>전체 게시글</option>
            </select>
            <select class="select-sort" name="sort" onchange="thi
                <option value="new" <%= "new".equals(sort) ? "sel
                <option value="old" <%= "old".equals(sort) ? "sel
            </select>
            <div class="search-form">
                <input
                    type="text"
                    name="q"
                    class="search-input"
                    placeholder="제목, 내용, 작성자 검색..."
                    value="<%= esc(q) %>"
                />
```

```
        <button type="submit" class="btn-search" aria-lab
            <svg viewBox="0 0 24 24" fill="none" stroke-w
        </button>
    </div>
</form>

<% if (dbError) { %>
    <div class="db-error">
        <div class="title">데이터베이스에 연결할 수 없습니다
        <div class="desc"><%= esc(dbErrorMsg) %></div>
    </div>
<% } else { %>

<div class="table-wrap">
    <table>
        <thead>
            <tr>
                <th style="width:60px;">번호</th>
                <th class="col-title">제목</th>
                <th style="width:150px;" class="col-autho
                <th style="width:110px;">작성일</th>
                <th style="width:120px;">관리</th>
            </tr>
        </thead>
        <tbody>
        <%
        SimpleDateFormat sdf = new SimpleDateFormat("yyyy
        for (Object[] row : postRows) {
            int id = (Integer) row[0];
            String rawAuthor = (String) row[2];
            String title = esc((String) row[1]);
            String author = esc(rawAuthor);
            java.sql.Timestamp ts = (java.sql.Timestamp)
            String dateStr = ts != null ? sdf.format(ts)
        %>
            <tr>
                <td class="col-no"><%= id %></td>
                <td class="col-title" title="<%= title %>
                    <a href="view.jsp?id=<%= id %>"><%= t
                </td>
                <td class="col-author">
                    <span class="author-cell">
                        <span class="avatar" style="backg
                        <span class="author-name"><%= aut
                    </span>
                </td>
                <td class="col-date"><%= dateStr %></td>
                <td class="col-manage">
                    <a href="edit.jsp?id=<%= id %>">수정</
                </td>
            </tr>
```

```
            <%
            }
            if (postRows.isEmpty()) {
            %>
                <tr>
                    <td colspan="5">
                        <div class="empty-state">
                            <div class="title"><%= q.isEmpty(
                            <div class="desc">
                                <% if (q.isEmpty()) { %>
                                    첫 번째 게시글을 작성해보세
                                <% } else { %>
                                    '<%= esc(q) %>'에 대한 검색
                                <% } %>
                            </div>
                        </div>
                    </td>
                </tr>
            <%
            }
            %>
            </tbody>
        </table>
    </div>

    <%
    if (totalPages > 1) {
        int windowStart, windowEnd;
        if (totalPages <= 7) {
            windowStart = 1; windowEnd = totalPages;
        } else if (curPage <= 4) {
            windowStart = 1; windowEnd = 5;
        } else if (curPage >= totalPages - 3) {
            windowStart = totalPages - 4; windowEnd = totalPa
        } else {
            windowStart = curPage - 2; windowEnd = curPage +
        }
        String qSuffix = q.isEmpty() ? "" : ("&q=" + qEncoded
        qSuffix += "old".equals(sort) ? "&sort=old" : "";
    %>
    <div class="pagination">
        <a class="nav-arrow<%= curPage <= 1 ? " disabled" : "
        <a class="nav-arrow<%= curPage <= 1 ? " disabled" : "
        <%
        if (windowStart > 1) {
        %>
            <a href="index.jsp?page=1<%= qSuffix %>">1</a>
            <% if (windowStart > 2) { %><span class="ellipsis
        <%
        }
        for (int p = windowStart; p <= windowEnd; p++) {
```

```
                    if (p == curPage) {
                %>
                        <span class="current"><%= p %></span>
                <%
                    } else {
                %>
                        <a href="index.jsp?page=<%= p %><%= qSuffix %
                <%
                    }
                }
                if (windowEnd < totalPages) {
                    if (windowEnd < totalPages - 1) {
                %>
                        <span class="ellipsis">…</span>
                <%
                    }
                %>
                    <a href="index.jsp?page=<%= totalPages %><%= qSuf
                <%
                }
                %>
                <a class="nav-arrow<%= curPage >= totalPages ? " disa
            </div>
            <%
            }
            %>

            <% } %>
        </div>
        </main>
    </body>
    </html>
```

5. **vi write.jsp**

```
<%@ page language="java" contentType="text/html; charset=UTF-8" p
<%@ page import="java.sql.*" %>
<%@ page import="javax.sql.*" %>
<%@ page import="javax.naming.*" %>
<%!
    // 간단한 HTML 이스케이프 (XSS 방지)
    private String esc(String s) {
        if (s == null) return "";
        return s.replace("&", "&amp;").replace("<", "&lt;")
                .replace(">", "&gt;").replace("\"", "&quot;");
    }
%>
<%
```

```
// sypark(2026.06.22) write.jsp - index.jsp와 동일한 UI 톤으로 작성
request.setCharacterEncoding("UTF-8");
response.setCharacterEncoding("UTF-8");

String title = request.getParameter("title");
String author = request.getParameter("author");
String content = request.getParameter("content");
if (title == null) title = "";
if (author == null) author = "";
if (content == null) content = "";

boolean isPost = "POST".equalsIgnoreCase(request.getMethod());
String errorMsg = null;

if (isPost) {
    title = title.trim();
    author = author.trim();
    content = content.trim();

    if (title.isEmpty() || author.isEmpty() || content.isEmpty())
        errorMsg = "제목, 작성자, 내용을 모두 입력해주세요.";
    } else {
        Connection conn = null;
        PreparedStatement pstmt = null;
        try {
            Context initContext = new InitialContext();
            Context envContext = (Context) initContext.lookup("ja
            DataSource ds = (DataSource) envContext.lookup("jdbc/
            conn = ds.getConnection();

            String sql = "INSERT INTO posts (title, author, conte
            pstmt = conn.prepareStatement(sql);
            pstmt.setString(1, title);
            pstmt.setString(2, author);
            pstmt.setString(3, content);
            pstmt.executeUpdate();

            response.sendRedirect("index.jsp");
            return;
        } catch (Exception e) {
            errorMsg = "데이터베이스에 연결할 수 없습니다. (" + e.g
        } finally {
            if (pstmt != null) try { pstmt.close(); } catch (Exce
            if (conn != null) try { conn.close(); } catch (Except
        }
    }
}
%>
<!DOCTYPE html>
<html>
<head>
```

```
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-sc
<title>글쓰기 - AWS 3Tier 실습</title>

<link rel="preconnect" href="https://fonts.googleapis.com">
<link rel="preconnect" href="https://fonts.gstatic.com" cross
<link href="https://fonts.googleapis.com/css2?family=Noto+San

<style>
    :root {
        --accent: #ef5350;
        --accent-hover: #e23e3b;
        --text-primary: #18181b;
        --text-secondary: #6b7280;
        --text-muted: #9ca3af;
        --border: #e5e7eb;
        --bg-page: #ffffff;
        --bg-soft: #f9fafb;
        --error-bg: #fef2f2;
        --error-border: #fecaca;
        --error-text: #b91c1c;
    }

    * { margin: 0; padding: 0; box-sizing: border-box; }

    body {
        font-family: 'Noto Sans KR', -apple-system, BlinkMacS
        background: var(--bg-page);
        color: var(--text-primary);
        padding: 48px 24px 80px;
    }

    .wrap { max-width: 720px; margin: 0 auto; }

    .page-header {
        display: flex;
        justify-content: space-between;
        align-items: flex-start;
        margin-bottom: 28px;
        gap: 16px;
        flex-wrap: wrap;
    }

    .page-title { font-size: 26px; font-weight: 700; margin-b

    .page-subtitle { font-size: 14px; color: var(--text-secon

    .btn {
        display: inline-flex;
        align-items: center;
        gap: 6px;
```

```
    padding: 10px 18px;
    border-radius: 8px;
    font-size: 14px;
    font-weight: 600;
    text-decoration: none;
    white-space: nowrap;
    border: none;
    cursor: pointer;
    font-family: inherit;
    transition: background .15s ease, border-color .15s e
}

.btn-primary { background: var(--accent); color: #fff; }
.btn-primary:hover { background: var(--accent-hover); }

.btn-secondary {
    background: #fff;
    color: var(--text-secondary);
    border: 1px solid var(--border);
}
.btn-secondary:hover { background: var(--bg-soft); }

.form-error {
    background: var(--error-bg);
    border: 1px solid var(--error-border);
    color: var(--error-text);
    border-radius: 10px;
    padding: 16px 20px;
    margin-bottom: 20px;
    font-size: 14px;
}

.form-card {
    border: 1px solid var(--border);
    border-radius: 10px;
    padding: 32px;
}

.field { margin-bottom: 22px; }
.field:last-of-type { margin-bottom: 0; }

.field label {
    display: block;
    font-size: 13px;
    font-weight: 600;
    color: var(--text-secondary);
    margin-bottom: 8px;
}

.field label .required { color: var(--error-text); margin
```

```
        .field input[type="text"],
        .field textarea {
            width: 100%;
            border: 1px solid var(--border);
            border-radius: 8px;
            padding: 11px 14px;
            font-size: 14px;
            font-family: inherit;
            color: var(--text-primary);
            background: #fff;
        }

        .field input[type="text"]:focus,
        .field textarea:focus {
            outline: none;
            border-color: var(--accent);
        }

        .field textarea {
            resize: vertical;
            min-height: 280px;
            line-height: 1.6;
        }

        .row-half {
            display: flex;
            gap: 16px;
        }

        .row-half .field { flex: 1; }

        .form-actions {
            display: flex;
            justify-content: flex-end;
            gap: 8px;
            margin-top: 28px;
            padding-top: 24px;
            border-top: 1px solid var(--border);
        }

        @media (max-width: 560px) {
            body { padding: 28px 16px 60px; }
            .page-title { font-size: 22px; }
            .form-card { padding: 22px; }
            .row-half { flex-direction: column; gap: 0; }
        }
    </style>
</head>
<body>
    <div class="wrap">
        <div class="page-header">
```

```
            <div>
                <div class="page-title">글쓰기</div>
                <div class="page-subtitle">AWS 3Tier 실습 게시판에
            </div>
            <a href="index.jsp" class="btn btn-secondary">목록으로
        </div>

        <% if (errorMsg != null) { %>
            <div class="form-error"><%= esc(errorMsg) %></div>
        <% } %>

        <form class="form-card" action="write.jsp" method="post">
            <div class="row-half">
                <div class="field">
                    <label for="author">작성자<span class="require
                    <input type="text" id="author" name="author" 
                </div>
            </div>
            <div class="field">
                <label for="title">제목<span class="required">*</s
                <input type="text" id="title" name="title" value=
            </div>
            <div class="field">
                <label for="content">내용<span class="required">*
                <textarea id="content" name="content" placeholder
            </div>

            <div class="form-actions">
                <a href="index.jsp" class="btn btn-secondary">취소
                <button type="submit" class="btn btn-primary">등록
            </div>
        </form>
    </div>
</body>
</html>
```

5개 jsp 파일을 Bastion-Server에 생성후 결과 화면



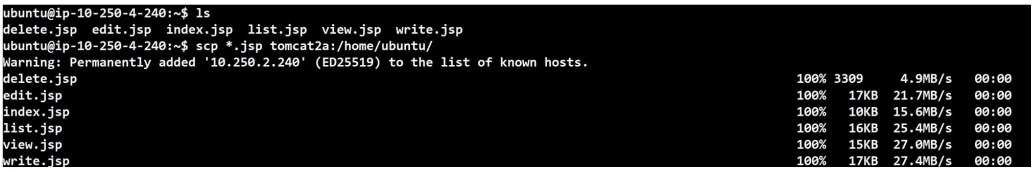
내려받은 파일을 tomcat2a와 tomcat2c로 옮기기

간편버전 명령어

```
scp *.jsp tomcat2a:/home/ubuntu/
```

원래버전 명령어

```
scp -i powermvp.pem *.jsp ubuntu@10.250.2.240:/home/ubuntu/
```



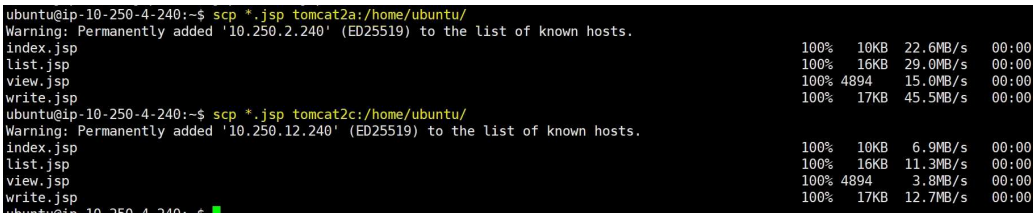
간편버전 명령어

```
scp *.jsp tomcat2c:/home/ubuntu/
```

원래버전 명령어

```
scp -i powermvp.pem *.jsp ubuntu@10.250.12.240:/home/ubuntu/
```





간편버전 명령어

```
ssh tomcat2a
```

원래버전 명령어

```
ssh -i powermvp.pem ubuntu@10.250.2.240
```

```
sudo -i
```

```
cd /home/ubuntu
```

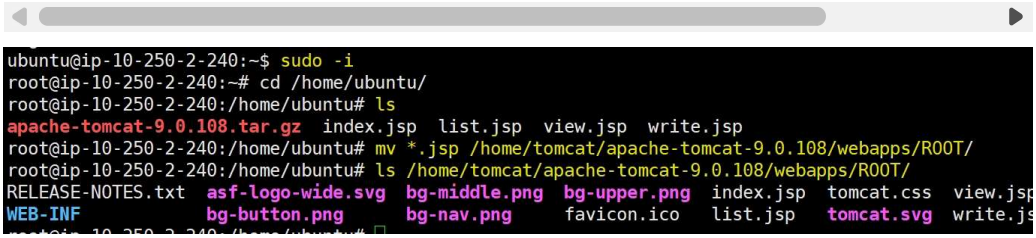
```
mv *.jsp /home/tomcat/apache-tomcat-9.0.108/webapps/ROOT/
```




```
ls /home/tomcat/apache-tomcat-9.0.108/webapps/ROOT/
```




```
sudo chown tomcat:tomcat /home/tomcat/apache-tomcat-9.0.108/webap
```



간편버전 명령어

```
ssh tomcat2c
```

원래버전 명령어

```
ssh -i powermvp.pem ubuntu@10.250.12.240
```

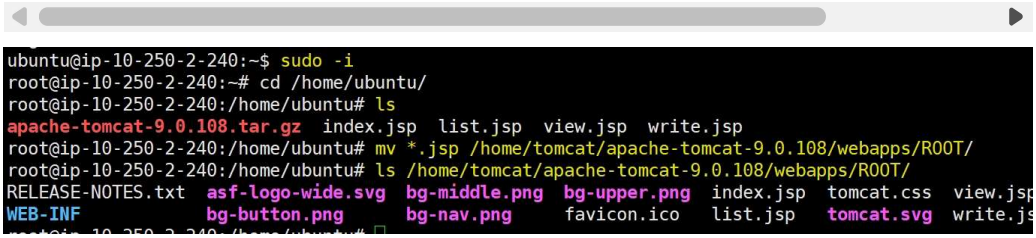
```
sudo -i
```

```
cd /home/ubuntu
```

```
mv *.jsp /home/tomcat/apache-tomcat-9.0.108/webapps/ROOT/
```

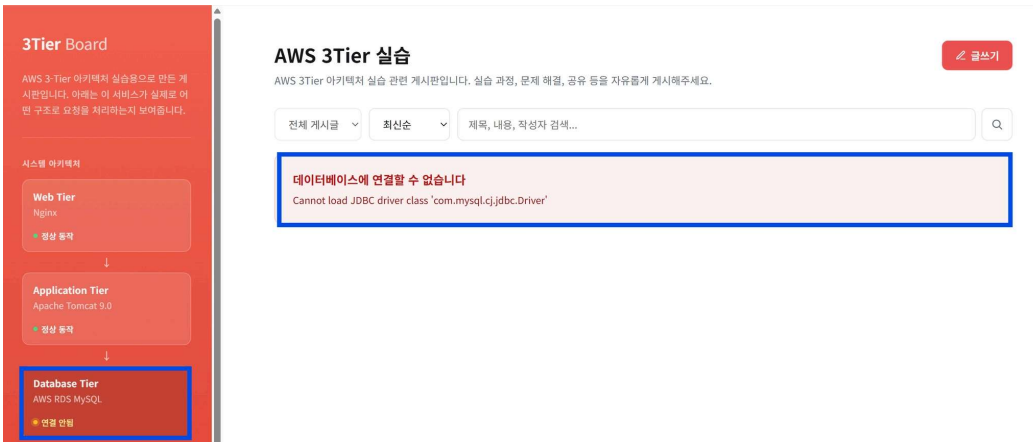
```
ls /home/tomcat/apache-tomcat-9.0.108/webapps/ROOT/
```

```
sudo chown tomcat:tomcat /home/tomcat/apache-tomcat-9.0.108/webap
```



ALB DNS 이름으로 접속 시 다음과 같이 새로고침해도 잘 나온다

다만 DB 연결이 안되었으므로 **데이터베이스 연결할 수 없습니다**. 에러 메세지 발생



위 문제를 해결하기 해 11. RDS 구성작업을 통해 문제를 해결해 보자. 11. RDS 구성작업 **편으로 이동해 봅시다.**